

# When Transparency Backfires: Strategic Obfuscation and Financial Fragility\*

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## Abstract

Abstract here

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# 1 Introduction

Obfuscation is common in investor-directed fund communications. Even when pursuing simple strategies, fund managers describe them in markedly different and often unnecessarily complex ways. For example, as [deHaan, Song, Xie, and Zhu \(2021\)](#) document, two funds that both track the S&P 500 describe essentially identical objectives with strikingly different clarity: Schwab states in a single sentence, “The fund’s goal is to track the total return of the S&P 500 Index,” while Deutsche describes it in 60 words, embedding the same objective in technical language.<sup>1</sup> Similar patterns arise in fund names: labels such as “Dynamic Opportunity Fund,” “Flexible Growth Strategy,” or “Opportunistic Equity Fund” convey little concrete meaning to unprofessional retail investors at first glance, yet in practice often correspond to relatively simple large-cap portfolios that closely resemble the S&P 500. This pattern is puzzling, as standard signaling theory suggests that managers, seeking to convey skill and attract capital, should prefer greater transparency rather than opacity. Why would managers make information harder for their own investors to interpret, and what are the incentives and market consequences when fund opacity becomes a manager’s strategic choice?

To address these questions, we study an equilibrium model of financial market trading with delegated asset management. A fund manager, privately informed about her skill, raises capital from investors through obfuscated communications and trades a risky asset in the financial market à la [Kyle \(1985\)](#). We define skill as the manager-specific ability to uncover more information about asset payoffs than the market: the more skilled the manager, the larger the mispricing she can identify and the greater the trading profits she can potentially earn. To attract fund flows, the manager privately communicates information about her risky position (portfolio) to investors. Investors, in

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<sup>1</sup>Deutsche’s S&P 500 index fund is described as “The fund seeks to provide investment results that, before expenses, correspond to the total return of common stocks publicly traded in the United States, as represented by the Standard & Poor’s 500 Composite Stock Price Index (S&P 500 Index). The fund invests for capital appreciation, not income; any dividend and interest income is incidental to the pursuit of its objective.”

turn, use this fund information to infer the manager’s skill and decide on capital allocation to the fund. Importantly, the manager strategically chooses the readability of this communication: by obfuscating information, she makes it more difficult for investors to back out her underlying skill.<sup>2</sup>

The model yields two stable, self-fulfilling equilibria regarding the manager’s trading strategy. One of the equilibria is similar to that of [Kyle \(1985\)](#), where the manager employs the profit-maximizing trading strategy. In the other, which we call the “flow-driven” equilibrium, the manager trades the risky asset more aggressively to appear skilled and to attract large fund flows. This strategy increases trading volume, price volatility, and market liquidity but undermines the fund expected performance by revealing too much private information to the market maker. Intuitively, the manager can boost investors’ perception of her skill by placing a large order, either buy or sell, because such an action makes her appear like a skilled manager who has identified an asset with significant mispricing. Although the expected performance deteriorates, it triggers large fund flows, and the manager earns a large fee income. Key to our multiple equilibria is the manager’s conflicting motives for revealing skill: on one hand, she wishes to hide her skill from the market maker to earn trading profits, but on the other hand, she wants to show it off to investors to establish high reputation and attract large fund flows.

The relative strength of these hiding and showing-off motives is governed by how sensitive fund flows are to the information the manager communicates about her portfolio, making the degree of fund opacity a key parameter determining the type of equilibrium. For example, when fund information is highly obfuscated, investors find it difficult to infer skill from fund information, and capital allocations respond only weakly to the manager’s trading strategy. In this case, the showing-off motive is weak, and the manager optimally follows the hiding motive, leading to a unique Kyle-like equilibrium. By contrast, when fund information is highly transparent, investors rely heav-

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<sup>2</sup>Following [Ellison and Ellison \(2009\)](#), the term “obfuscation” refers to general practices that reduce the readability of information and distort communications. Accordingly, “transparency” and “opacity” refer to the degree of interpretability and precision of information with which investors can assess the fund’s positions, rather than mere disclosure.

ily on it in estimating the manager’s skill. As fund flows become sensitive to the manager’s trading strategy, the manager finds it beneficial to inflate risky positions for large flows, so that a unique flow-driven equilibrium with excessive risk-taking arises. At intermediate levels of opacity, these competing forces are balanced, giving rise to multiple equilibria. Importantly, these equilibria are self-fulfilling and depend on investors’ beliefs about manager’s trading behavior. If investors, for whatever reasons, believe that the manager is trading aggressively at the flow-driven equilibrium level, they find the fund information an informative signal about the manager’s skill. Fund flows grow sensitive to the manager’s trading strategy, and the showing-off motive is amplified. Consequently, the manager conforms to such a belief and indeed trades aggressively. The symmetric argument holds when investors believe that the manager takes a solid Kyle-like trading strategy. The self-fulfilling property of our equilibria implies that a mere shift in investor beliefs, unaccompanied by changes in fundamentals, can trigger a transition from the standard Kyle-like equilibrium to the flow-driven one, where price volatility is high and trading performance is relatively poor. We interpret this as *fragility* in financial markets.<sup>3</sup>

The equilibrium multiplicity in trading styles has implications for the fund-management industry. For example, the flow-driven equilibrium explains observations by Philippon and Reshef (2012) and C  lerier and Vall  e (2019), who document increasing returns to talent and their significant contribution to income inequality in the finance industry. In the flow-driven equilibrium, managers’ fee income exhibits greater inequality among high- and low-skilled managers compared to those in the Kyle-like equilibrium, as they trade aggressively and reveal their skills more than they do in the Kyle-like equilibrium. This allows investors to better assess each manager’s skill and, therefore, to allocate capital more accurately according to managers’ skills. Moreover, multiple equilibria offer a novel explanation for the puzzling relationship (or the lack thereof) between skill and persistent performance. In our model, skill

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<sup>3</sup>As in Greenwood and Thesmar (2011), we define a market as fragile if it is susceptible to non-fundamental demand shifts, such as those caused by changes in investor beliefs.

does not translate into persistent performance, as high-skilled managers in the flow-driven equilibrium may underperform low-skilled ones following the Kyle-like equilibrium, consistent with the evidence documented in the empirical literature (e.g., [Busse, Goyal, and Wahal, 2010](#); [Berk and Green, 2004](#)).

Building on the tension between hiding and showing-off motives and the resulting multiple equilibria, we consider managers’ strategic incentives to obfuscate fund information and analyze for what types of managers financial fragility becomes a concern. By obfuscating fund information, a manager weakens investors’ responsiveness to reported positions and curbs her own showing-off incentive at the portfolio-decision stage. Although, when deciding on the risky position, she takes investors’ belief-updating rule as given and is tempted to over-trade to attract flows, *ex-ante* she can influence investors’ belief-updating rule through obfuscation. In this sense, opacity serves as a commitment device: by tying her own hands, she mitigates inefficient risk-taking at the portfolio-decision stage. It raises the profit margin of her risky position and the fund’s expected performance, thereby benefiting both herself and investors.

The obfuscation incentive is stronger for managers with higher expected skill than those with low expected skills, as the former type of managers benefit more from the improved profit margin arising from obfuscation. This effect shapes the optimal degree of obfuscation into an increasing function of managerial skills, offering a unified interpretation of the observed relationship among skill, opacity, and trading style in practice. For instance, our results predict that high-skilled managers strategically operate with substantial opacity and trade under the unique Kyle-like equilibrium, focusing on performance rather than reputation-driven fund flows. It is somewhat consistent with the “brain drain” documented by [Kostovetsky \(2017\)](#), whereby top mutual fund managers leave to start more opaque hedge funds and deliver superior performance. In contrast, managers of intermediate skill prefer intermediate opacity, which exposes them to belief-driven multiple equilibria. Accordingly, they contribute to market fragility, as they switch between solid profit-seeking strategies and flashy flow-driven risk taking based on market beliefs. Such behaviors are

consistent with momentum crashes or sudden reversals often observed during financial crises (Brunnermeier, 2009), even in the absence of fundamental shocks, and our model attributes them to intermediate-skilled managers operating under intermediate opacity.

Our paper contributes to the understanding of opacity, complexity, and obfuscation in financial markets. Empirically, Edelen, Evans, and Kadlec (2012), Badoera, Costello, and James (2020), and deHaan, Song, Xie, and Zhu (2021) document that mutual funds strategically obfuscate disclosures of their strategies and fee structures, and C  lerier and Vall  e (2019) show that structured financial products are deliberately made complex. Building on such evidence, Carlin (2009), Carlin and Manso (2010), and Sato (2014) develop theoretical models in which complexity enables financial institutions to extract fees or preserve informational advantages. In these models, however, opacity is typically treated as technologically given or engineered externally, while we analyze its implications when fund managers strategically obfuscate. Moreover, opacity in our framework improves both managers' and investors' *ex-ante* utility. Unlike the prior literature, managers obfuscate not to exploit less-informed investors, but because opacity serves as a commitment device that mitigates managers' showing-off motive in financial markets.

Our paper is also related to the literature on fragility and trading frenzies in financial markets. Cespa and Vives (2015) is most closely related, as they also find multiple equilibria arising from short-term investor horizons and persistent liquidity trading. In Froot, Scharfstein, and Stein (1992), strategic complementarities among short-term traders are also important for the emergence of multiple equilibria. Unlike these studies, our model does not feature short horizons; instead, we obtain fragility from managers' concerns about fund flows. In Cespa and Foucault (2014), learning from other assets amplifies shocks and causes liquidity crashes. Although the mechanism is different, our model can also result in liquidity crashes when the economy switches between equilibria due to non-fundamental shocks. Also, several studies, such as Veldkamp (2006) and Goldstein, Ozdenoren, and Yuan (2013), feature trading frenzies based on strategic complementarities in information markets.

or a feedback effect from financial markets to a firm’s real investment. By contrast, in our flow-driven equilibrium, a fund manager trades aggressively out of fear of being perceived as less skilled by the recruiter who believes that she would trade aggressively.

Lastly, our work contributes to the broader understanding of the role of asymmetric information about skills in financial markets. [Huberman and Kandel \(1993\)](#), [Huddart \(1999\)](#), [DiMaggio \(2015\)](#), [Malliaris and Yan \(2021\)](#), and [Bijlsma, Boone, and Zwart \(2018\)](#) derive implications for the risk-taking behavior of agents when ability is private information but is conveyed to investors through signals, such as realized performance. In the context of the principal-agent problem, [Prat \(2005\)](#) shows that transparency in such communications can be detrimental, as it distorts agents’ behavior. We also link trader skill to private information and asset markets; however, unlike these papers, our focus is on deriving pricing implications and managers’ incentive to obfuscate fund information. [Guerrieri and Kondor \(2012\)](#) show that career concerns amplify price volatility, similar to our findings. In [Dasgupta \(2006\)](#), career concerns lead to churning (trading without private information), and in [Dasgupta and Prat \(2008\)](#), they lead to herding (ignoring private information and trading like everyone else). These three papers are closely related to ours in that they study career concerns and endogenous asset prices, but they do not analyze fragility and endogenous fund opacity, the main focus of our paper. [Gervais and Strobl \(2020\)](#) consider managers’ transparency choices under exogenous transparency costs and obtain sorting across skill types. By contrast, we derive an endogenous cost of transparency arising from flow-driven trading distortions. As a result, transparency in our model does not merely separate types but alters trading behavior and can generate belief-driven multiple equilibria and market fragility.

The rest of the paper proceeds as follows. Section 2 presents a baseline model with exogenous reputation concerns, and Section 3 studies the equilibrium. Section ?? endogenizes reputation concerns by introducing a two-period model and presents the implications of our results. Section ?? analyzes traders’ skill acquisition. The [Online Appendix](#) contains all proofs for the theoretical

results.

## 2 Model

This section presents a one-period trading model à la Kyle (1985) with delegated asset management.

### 2.1 Setting

The economy consists of a *fund manager*, a competitive *market maker*, a *noise trader*, and  $n \geq 2$  of *investors*. All players are risk neutral. The manager leverages her skill (defined below) to identify a single risky asset with potential mispricing, raises capital from investors, and allocates it between the risky asset and a risk-free asset with a zero interest rate.<sup>4</sup>

#### 2.1.1 Skill

The manager can uncover more information about the asset's payoff than the market maker, and we interpret it as her *skill*. At the beginning of the period, the manager draws private information,  $s$ , from a normal distribution with mean zero and variance  $\omega_s$ . Leveraging this information, she identifies an asset with payoff  $\delta = \bar{\delta} + s$ , where  $\bar{\delta}$  is a publicly known mean payoff. Since the market maker does not observe  $s$ , her prior expectation of the asset's payoff is  $E[\delta] = \bar{\delta} + E[s] = \bar{\delta}$ , which is biased by  $s$  from the informed manager's perspective. As the market maker would set a price  $p = E[\delta] = \bar{\delta}$  in the absence of trades,  $s$  can be viewed as the asset's "potential mispricing."<sup>5</sup> Indeed, as shown later, the manager profits by buying the asset if  $s > 0$  (underpriced) and selling it if  $s < 0$  (overpriced); the larger the  $|s|$ , the larger the profit.

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<sup>4</sup>We abstract away from information acquisition by individual investors and assume that they invest through a manager due to the lack of informational advantages. This assumption would be supported by, for example, information- or skill-acquisition costs, where only the fund manager acquires the skill at lower costs due to economies of scale (e.g., Gârleanu and Pedersen, 2018).

<sup>5</sup> $s$  is not the *actual* mispricing because the market maker will take into consideration the presence of the informed manager; thus,  $p$  will deviate from  $\bar{\delta}$  in equilibrium.



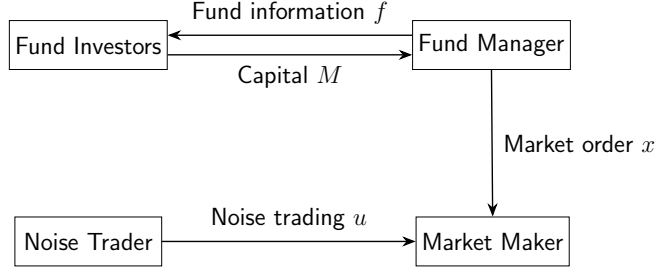


Figure 1: Capital Allocation and Investment

In light of this, we measure the manager’s *realized* asset-selection skill by  $|s|$ , while the variance  $\omega_s$  serves as an *ex-ante* measure of the average skill.<sup>6,7</sup>

### 2.1.2 Capital Allocation and Trading

We model the manager’s portfolio choice and investors’ capital allocation as a simultaneous game with rational expectations, as illustrated in Figure 1.

*Manager’s portfolio choice.* Upon discovering the mispricing in the asset ( $s$ ), the fund manager chooses the portfolio allocation between the risky and the safe asset. This action is represented by a market order for  $x \in \mathbb{R}$  units of the risky asset, submitted to the market maker in the financial market. The manager decides on  $x$  by incorporating the reactions of the financial market (i.e., the asset price) and the capital inflow from her investors, as described in the following.

*Investors’ capital allocation.* Each investor chooses the amount of capital to invest into the fund. Investor  $i$ ’s capital allocation is denoted by  $m_i \geq 0$ , and the total fund flow or the asset under management (AUM) of the fund is defined by  $M \equiv \sum_{i=1}^n m_i$ . The fund’s fee structure is proportional to the total

<sup>6</sup>This formalization of skill aligns with findings in the empirical studies that attribute fund managers’ performance to stock-picking skills (e.g., Wermers, 2000; Kosowski, Timmermann, Wermers, and White, 2006).

<sup>7</sup>The definition of the average skill follows from  $E[|s|] = \omega_s \sqrt{\frac{2}{\pi_0}}$ , where  $\pi_0$  denotes the circular constant.

fund proceeds: the manager takes a  $\phi \in (0, 1)$  fraction of the proceeds, while the remaining  $1 - \phi$  fraction is distributed to investors in proportion to their contribution to the fund, that is,  $m_i/M$ .<sup>8</sup>

*Financial market.* In the financial market, the market maker sets the asset price,  $p$ , based on the incoming order flow,  $x + u$ , where  $u$  represents a random market order from the noise trader and follows a normal distribution with mean zero and variance  $\omega_u$ . Due to competition, the market maker sets a semi-strong efficient price conditional on  $x + u$ .<sup>9</sup>

$$p = E[\delta | x + u]. \quad (2.1)$$

### 2.1.3 Obfuscation and Fund Opacity

At the fund-raising stage, the manager disseminates a private noisy signal about her trading strategy to investors, defined as  $f = x + \eta$ , where  $\eta$  represents a normally distributed noise term with mean zero and variance  $\omega_\eta$ .<sup>10</sup> This signal is referred to as the *fund information*, and the noise variance of fund information,  $\omega_\eta$ , is used as the measure of *fund opacity*. It describes intentional obfuscation of the information that investors receive about the manager's true trading strategy. A higher  $\omega_\eta$  means that the fund information is more complex, making the strategy harder to verify or reverse engineer. This reflects the idea that managers may deliberately obfuscate details by being vague, selective, or technically complex in their descriptions, while still engaging with investors to raise capital and to comply with disclosure rules. Since the manager chooses her trading strategy  $x$  based on the private signal  $s$ , the fund

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<sup>8</sup>Introducing a fixed fee does not change our qualitative results, as long as the performance-based fee also exists,  $0 < \phi < 1$ .

<sup>9</sup>Assume that, on the equilibrium path, one competitive market maker trades the asset. As in Kyle (1985), competitively many market makers exist off the equilibrium path, ensuring a break-even price in the equilibrium.

<sup>10</sup>We assume that  $f$  is privately communicated between the manager and investors and is not observable to the market maker. Allowing the market maker to observe a noisy signal about  $x$ , however, does not affect the main qualitative results, as long as this signal is conditionally independent of  $f$ .

information  $f$  is informative about the manager's skill and the fund's return. Thus, fund investors use  $f$  to decide on their capital allocations. In what follows, we first consider the baseline model with exogenous opacity, while Section 4 endogenizes  $\omega_\eta$  by allowing the manager to strategically obfuscate.

#### 2.1.4 Maximization Problems and Equilibrium

The fund's profit from the risky investment is represented by  $\pi \equiv (\delta - p)x$ , so that the total fund proceeds, including those from the safe asset, are

$$y = \pi + M. \quad (2.2)$$

Therefore, given the realized skill  $s$ , the fund manager maximizes her expected fee income by controlling her investment strategy  $x$ :

$$\max_x \mathbb{E}[\phi y | s]. \quad (2.3)$$

Conversely, conditional on the fund information  $f$ , investor  $i$  maximizes her expected return from delegating capital management:<sup>11</sup>

$$\max_{m_i \geq 0} \mathbb{E} \left[ \frac{m_i}{M} (1 - \phi) y - m_i | f \right]. \quad (2.4)$$

Note that the first term represents the expected payment from the fund, and the second term is the cost of capital.

**Definition 1.** *The equilibrium is defined by the market maker's pricing strategy  $p$ , the fund manager's trading strategy  $x$ , and investors' capital allocations  $\{m_i\}_{i=1}^n$ , such that, (i)  $p$  satisfies (2.1) given  $x + u$ , (ii)  $x$  solves (2.3) given  $s$ ,  $p$ , and  $\{m_i\}_{i=1}^n$ , and (iii) for all  $i$ ,  $m_i$  solves (2.4) given  $p$ ,  $f$ , and  $\{m_j\}_{j \neq i}$ .*

<sup>11</sup>The fund manager may have an incentive to make a false report about  $y$  to her investors. Namely, by reporting  $\tilde{y}$ , she pays out  $(1 - \phi)\tilde{y}$  to investors and captures  $y - (1 - \phi)\tilde{y}$  on her own. To prevent this intentional misreporting, we assume that each investor can verify  $x$  after the trading stage. Alternatively, we may assume costly verification: investors collectively pay the verification cost  $C$  to back out  $x$  from the fund information in the end of the trading period. As long as  $C$  positively depends on the degree of fund opacity  $\omega_\eta$ , our qualitative results remain the same.

### 3 Equilibrium

In this section, we search for a linear equilibrium in which the manager's market order  $x$  is linear in her skill  $s$ , and the market maker's pricing strategy  $p$  is also linear in the order flow  $x + u$ . Proofs for the analytical results are provided in the [Appendix](#).

#### 3.1 Trading Strategy

Conjecture and later verify that the manager's equilibrium trading strategy is

$$x(s) = \beta s, \quad (3.1)$$

where  $\beta > 0$  is an endogenous constant determined in the equilibrium. Conjecture (3.1) states that the manager buys (sells) the asset if her  $s$  is positive (negative), where the order size is proportional to her skill  $|s|$ ; the more skilled the manager, the more aggressively she trades.<sup>12</sup> The scale factor  $\beta$  is referred to as the *trading intensity*, representing how intensively the manager exploits her informational advantage over the market maker.

#### 3.2 Execution Price

The market maker decides on the price of the asset by extracting information about the asset's payoff from the aggregate order flow  $x + u$ . Anticipating the trading strategy (3.1), the price in equation (2.1) can be rewritten as

$$p = \bar{\delta} + \lambda(x + u), \quad (3.2)$$

where

$$\lambda \equiv \frac{\beta \omega_s}{\beta^2 \omega_s + \omega_u} \quad (3.3)$$

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<sup>12</sup>This conjecture follows from the original Kyle model where an informed trader trades based on the difference between her belief and that of the market maker, i.e.,  $E[\delta|s] - E[\delta] = s$ .

is referred to as “Kyle’s lambda,” a measure of the price impact of order flow.<sup>13</sup> Following the literature, we define a market to be *liquid* if  $\lambda$  is small.

### 3.3 Investors’ Capital Allocation

By incorporating the fund proceeds  $y = \pi + M$ , investor  $i$ ’s optimization problem in (2.4) is

$$\max_{m_i \geq 0} \frac{m_i}{M} (1 - \phi) \mathbb{E}[\pi|f] - \phi m_i. \quad (3.4)$$

Investor  $i$  solves this problem by incorporating the impact of her choice ( $m_i$ ) on the total fund flow ( $M$ ), while taking other investors’ behavior ( $\sum_{j \neq i} m_j$ ) as given. As she allocates a larger amount of capital, its marginal return diminishes due to the pro-rata allocation of the fund’s proceeds, while the cost of capital linearly increases. This structure ensures the second-order condition of problem (3.4), and the first-order condition pins down the optimal capital allocation as follows:

$$m_i = \left( \frac{1 - \phi}{\phi} \mathbb{E}[\pi|f] \sum_{j \neq i} m_j \right)^{\frac{1}{2}} - \sum_{j \neq i} m_j. \quad (3.5)$$

We focus on the symmetric equilibrium where all investors take the same action,  $m_i = m$  for all  $i = 1, 2, \dots, n$ . In this equilibrium, equation (3.5) reduces to

$$m = \frac{n-1}{n^2} \frac{1 - \phi}{\phi} \mathbb{E}[\pi|f], \quad (3.6)$$

and the total fund flow becomes

$$M \equiv nm = \theta \mathbb{E}[\pi|f], \quad (3.7)$$

where  $\theta \equiv \frac{n-1}{n} \frac{1-\phi}{\phi}$  represents the *flow-performance sensitivity*. Both the individual capital allocation and the total fund flow are linearly increasing in

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<sup>13</sup>Since the market maker believes that the manager follows (3.1), the linear filtering rule with normally distributed random variables yields  $p = \mathbb{E}[\bar{\delta} + s|\beta s + u] = \bar{\delta} + \mathbb{E}[s] + \frac{\text{Cov}[s, \beta s + u]}{\text{Var}[\beta s + u]}(x + u - \mathbb{E}[\beta s + u]) = \bar{\delta} + \frac{\beta \omega_s}{\beta^2 \omega_s + \omega_u}(x + u)$ .

the expected trading profit conditional on the fund information,  $E[\pi|f]$ . This is due to the performance-based fee structure, which makes the per-unit return from delegated asset management increasing in the trading profit. Also, equation (3.7) specifies the flow-performance sensitivity  $\theta$ , which governs the reaction of the fund flow to the expected fund performance. The fund flow becomes less sensitive to the expected performance when the fee rate ( $\phi$ ) is high, as high  $\phi$  raises the cost of delegation and discourages investment, while it becomes more sensitive when the fund has a large clientele ( $n$ ), as it attracts capital from a larger number of investors.

### 3.4 Reputation

In determining the capital allocation in (3.6), each investor seeks to compute the expected fund profits  $E[\pi|f]$ . This process, in turn, requires the investor to back out the realized skill level of the manager  $|s|$  from fund information  $f = x + \eta$ .

**Lemma 3.1.** *Upon observing the fund information  $f = x + \eta$ , investors update their expectation about the trading profit as*

$$E[\pi|f] = (1 - \lambda\beta)\beta E[s^2|f] = (1 - \lambda\beta)\beta (\hat{s}^2 + \hat{\omega}_s), \quad (3.8)$$

where the posterior expectation of the (signed) realized skill is

$$\hat{s} = E[s|f] = \xi(x + \eta), \quad (3.9)$$

with

$$\xi = \frac{\beta\omega_s}{\beta^2\omega_s + \omega_\eta}, \quad (3.10)$$

and the posterior of the skill variance is

$$\hat{\omega}_s = \text{Var}[s|f] = \frac{\omega_s\omega_\eta}{\beta^2\omega_s + \omega_\eta}. \quad (3.11)$$

Equation (3.8) suggests that the fund return,  $\pi = (\delta - p)x$ , becomes a

quadratic function of  $s$ , as both the manager's trading quantity  $x$  and the profit margin,  $\delta - p$ , are proportional to  $s$  in expectation. Therefore, investors must update their belief about the manager's skill based on fund information. Equation (3.9) describes how investors handle this updating process:  $\hat{s}$  represents the updated expectation about skill and is referred to as the *reputation* in the following discussion. According to equation (3.9),  $\xi$  measures how much investors rely on  $f$  in updating their belief and governs the sensitivity of the reputation to the fund's trading strategy  $x$ . Holding  $\beta$  constant,  $\xi$  is large when investors' prior is unreliable ( $\omega_s$  is large) and the fund information is transparent ( $\omega_\eta$  is small). More importantly,  $\xi$  itself also depends on the trading strategy  $\beta$ , as a more intensive trading strategy makes  $f$  more informative about  $s$ .

Lemma 3.1 implies that, on top of the maximizing profits, delegated asset management generates an additional incentive for the manager to control her trading strategy  $x$ , as it affects the total fund flow  $M$  by influencing the investors' expectation about fund's returns.

### 3.5 Manager's Optimization

By incorporating the total fund flow ([3.7] and [3.8]), the manager's problem in (2.3) is re-written as follows:

$$\max_x sx - \lambda x^2 + \theta(1 - \lambda\beta)\beta [\xi^2(x^2 + \omega_\eta) + \hat{\omega}_s]. \quad (3.12)$$

The first two terms represent the fee income from the investment return and are essentially identical to the trading profit in the original Kyle (1985) model. The last term with coefficient  $\theta$  arises from the fee income associated with the fund flow  $M$ . It increases with the manager's risk taking, measured by  $x^2$ , because she can potentially inflate her reputation  $|\hat{s}|$  by placing a large order (either buy or sell). The reaction of the expected fund flow to the trading strategy is influenced not only by the flow-performance sensitivity  $\theta$  but also by the investors' updating coefficient  $\xi$ . This is because fund opacity ( $\omega_\eta > 0$ ) prevents investors from perfectly inferring the manager's trading strategy from

fund information. Importantly, investors' estimate of the manager's skill and, therefore, the reaction of the fund flow to the risky investment depend on the risky trading strategy  $\beta$  itself: the more intensive the manager invests, the more reliable the fund information becomes for investors. This structure is captured by the dependence of  $\xi$  on  $\beta$  in equation (3.10) and leads to the strategic complementarity between the manager's risk taking and investors' capital allocation based on reputation.

The unique solution to the manager's optimization problem in (3.12) (given  $\beta$ ) is represented by

$$x = F(\beta)s, \quad (3.13)$$

where

$$F(\beta) \equiv \frac{1}{2(\lambda - \theta(1 - \lambda\beta)\beta\xi^2)}. \quad (3.14)$$

$F(\cdot)$  is a nonlinear function of  $\beta$  due to the dependence of  $\lambda$  and  $\xi$  on  $\beta$ .

### 3.6 Equilibrium under Opacity

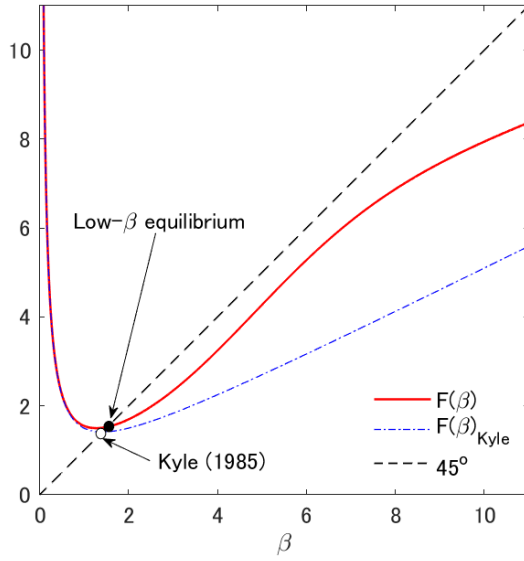
The manager optimally chooses (3.13) given that the market maker and investors believe that she follows strategy (3.1). In equilibrium, their beliefs must be correct, meaning that (3.1) and (3.13) must be consistent with each other. This belief consistency condition requires that

$$\beta = F(\beta). \quad (3.15)$$

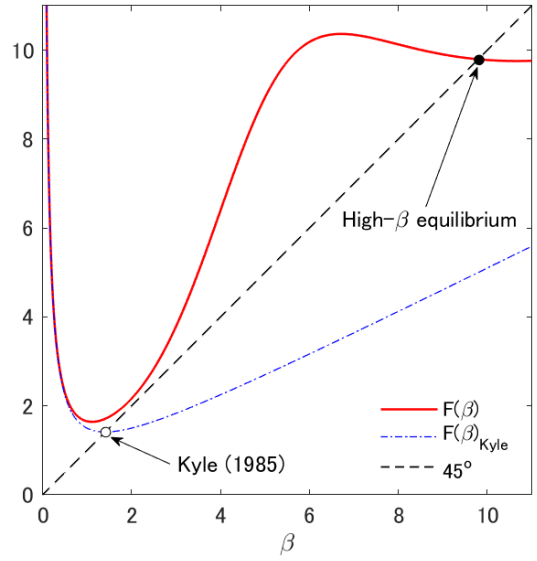
That is, the equilibrium levels of  $\beta$  are determined at the fixed points of  $F(\cdot)$ . Having identified  $\beta$ , the equilibrium levels of  $x$ ,  $p$ , and  $M$  are determined by equations (3.1), (3.2) and (3.7), respectively. In what follows, we first illustrate the determination of equilibrium  $\beta$  through a numerical example, followed by an analytical proposition that formally establishes the equilibria.

Figure 2 illustrates the determination of  $\beta$  for three different levels of fund opacity  $\omega_\eta$ . The intersections of  $F(\beta)$  (the solid line) and the 45-degree line yield the equilibrium levels of  $\beta$ . We also plot  $F(\beta)_{\text{Kyle}}$  in the dash-dotted line, representing the original Kyle (1985) model. This benchmark case arises when

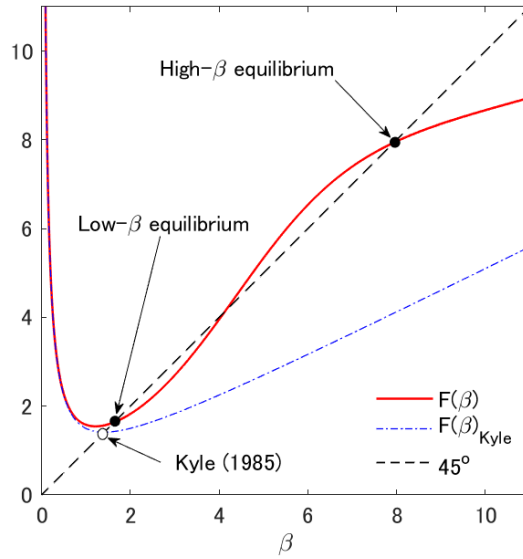




(a): Large  $\omega_\eta$  ( $\omega_\eta = 48$ )  $\Rightarrow$  unique low- $\beta$  equilibrium.



(b): Small  $\omega_\eta$  ( $\omega_\eta = 28$ )  $\Rightarrow$  unique high- $\beta$  equilibrium.



(c): Intermediate  $\omega_\eta$  ( $\omega_\eta = 38$ )  $\Rightarrow$  multiple high- $\beta$  & low- $\beta$  equilibria.

Figure 2: Determination of  $\beta$

Note: The figure plots  $F(\beta)$  for different values of  $\theta$ . The equilibrium values of  $\beta$  are determined at the fixed points of  $F(\cdot)$ . The parameter values used in the figures are  $\omega_s = 1$ ,  $\omega_u = 2$ , and  $\phi = 0.02$ .

the fund is infinitely opaque (i.e.,  $\omega_\eta \rightarrow \infty$ ), as it makes the fund flow inelastic to the manager’s trading strategy, and the delegated asset management becomes irrelevant.

Panel (a) of Figure 2 shows that if  $\omega_\eta$  is large, there is a unique “low- $\beta$  equilibrium,” in which  $\beta$  is close to the trading intensity without delegated asset management. Panel (b) shows that a large  $\omega_\eta$  leads to a unique “high- $\beta$  equilibrium,” in which  $\beta$  is much larger than that in the original Kyle (1985) model. Panel (c) shows that multiple equilibria arise for an intermediate level of  $\omega_\eta$ . Specifically, the high- $\beta$  and the low- $\beta$  equilibria are stable, as  $F(\beta)$  crosses the 45-degree line from above, whereas the one with intermediate  $\beta$  is unstable, as  $F(\beta)$  crosses the line from below.<sup>14</sup> In what follows, we focus only on the stable equilibria.

*Fund opacity and multiple equilibria.* To see the intuition behind the different equilibrium patterns presented in Figure 2, note that the fund manager faces two conflicting motives when trading. On the one hand, she seeks to avoid trading too aggressively on her private information, as doing so would move the price excessively and reduce her trading profits. As in Kyle (1985), this motive leads the manager to decrease  $\beta$ . On the other hand, she wishes to trade intensively to signal her skill to her investors, aiming to establish a strong reputation and attract large fund flows. This motive encourages her to increase  $\beta$ . In summary, she faces a tradeoff between “hiding” her skill from the market maker and “showing it off” to her investors. The fund opacity  $\omega_\eta$  governs this tradeoff by influencing the sensitivity of the manager’s reputation held by investors to her investment behavior. In particular, Lemma 3.1 demonstrates that the fund flow  $M$  becomes less sensitive to  $x$  under opacity (i.e.,  $\xi$  is small) compared to the fully transparent case, as it becomes difficult for investors to estimate the skill based on the fund information  $f$ . Therefore, fund opacity diminishes the strategic complementarity between the manager’s risk taking

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<sup>14</sup>We define an equilibrium to be stable (unstable) if the corresponding  $\beta$  is a stable (unstable) fixed point of  $F(\cdot)$ , that is,  $|F'(\beta)| < 1$  ( $> 1$ ). A stable equilibrium is robust to a small perturbation to the agents’ belief about  $\beta$ , meaning that the economy converges back to the original equilibrium through the best-response dynamics of the agents’ beliefs.

and the reputation-driven capital allocation by investors, thereby weakening the showing-off motive.

For an opaque fund with a large  $\omega_\eta$  (panel [a]), the hiding motive dominates the showing-off motive, leading to a unique low- $\beta$  equilibrium. In this equilibrium, investors believe that the manager’s trading intensity is relatively low and the fund information is not very informative, making the reputation and the fund flow insensitive to  $x$ . Although the manager could inflate her reputation by placing an order larger than what her investors anticipate, she refrains from doing so, because it would cause a large price impact and reduce the fund’s investment profits, while fund flows and her fee income would not increase much due to small  $\xi$ . Consequently, the equilibrium  $\beta$  is close to the trading intensity without delegated asset management, meaning that it is a “Kyle-like equilibrium.”

For a transparent fund with a small  $\omega_\eta$  (panel [b]), the showing-off motive prevails, resulting in a unique high- $\beta$  equilibrium. In this equilibrium, investors believe that the manager is trading aggressively. The reputation and fund flows grow sensitive to the manager’s trading strategy through the fund information. Under such beliefs, the manager is actually willing to trade aggressively: although it moves the price substantially and undermines investment profits, it helps her build a strong reputation and generates a large fee income from significant capital inflows. Importantly, investors’ learning is not manipulated, as their belief about the manager’s trading strategy is correct. Nonetheless, the manager places a large order because doing otherwise would induce her investors to incorrectly estimate that she is less skilled, thereby deteriorating her reputation. She trades aggressively solely to conform to her investors’ belief and to attract large fund flows. To emphasize this mechanism, we call this equilibrium the “flow-driven equilibrium.”

For intermediate levels of  $\omega_\eta$  (panel [c]), the strategic complementarity between investors’ reputation-based capital allocation and the manager’s risk taking is neither weak nor strong, and there is no clear dominance between the two motives. Consequently, both the low- $\beta$  and the high- $\beta$  equilibria are supported under intermediate fund opacity. Each equilibrium is associated

with a self-fulfilling belief of the market maker and investors.<sup>15</sup>

The following proposition summarizes the above discussion and provides the analytical characterization of the equilibria.

**Proposition 3.1** (Equilibrium characterization). *There exists a critical value of the flow-performance sensitivity, denoted as  $\theta_0$ .*

- (i) *If  $\theta \leq \theta_0$ , the equilibrium is unique and characterized by a threshold of fund opacity  $\omega_0$ , such that,  $\omega_\eta \leq \omega_0$  leads to the high- $\beta$  equilibrium and  $\omega_\eta \geq \omega_0$  leads to the low- $\beta$  equilibrium.*
- (ii) *If  $\theta > \theta_0$ , there are two thresholds of fund opacity,  $\omega_L$  and  $\omega_H (> \omega_L)$ . When  $\omega_\eta \leq \omega_L$ , the high- $\beta$  equilibrium is unique; when  $\omega_\eta \geq \omega_H$ , the low- $\beta$  equilibrium is unique; and when  $\omega_\eta \in (\omega_L, \omega_H)$ , the high- and low- $\beta$  equilibria coexist.*

Figure 3 illustrates the result in Proposition 3.1. It indicates that both fund opacity  $\omega_\eta$  and flow-performance sensitivity  $\theta$  are critical in determining the equilibrium type, as they jointly influence the reaction of the fund flow to changes in the manager's trading strategy, thereby affecting her showing-off motive. As high levels of  $\theta$  strengthen this motive, we obtain the following corollary:

**Corollary 3.1.** (i) *When  $\theta \leq \theta_0$ , the threshold of fund opacity  $\omega_0$  is monotonically increasing in  $\theta$  with  $\lim_{\theta \rightarrow 0} \omega_0 = 0$ .*

- (ii) *When  $\theta > \theta_0$ , both  $\omega_L$  and  $\omega_H$  are increasing in  $\theta$ . Moreover, the region for multiple equilibria,  $(\omega_L, \omega_H)$ , expands as  $\theta$  increases.*

For a given value of fund opacity  $\omega_\eta$ , a weak flow-performance sensitivity supports only the low- $\beta$  equilibrium, as the hiding motive strictly dominates the weak showing-off motive. However, as  $\theta$  increases, a high flow-performance sensitivity encourages the manager to show off her skill by trading intensively. Hence, it gives birth to the high- $\beta$  equilibrium. The tradeoff between the

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<sup>15</sup>For simplicity, we assume that agents do not agree to disagree, so that they coordinate their belief when parameter values are such that multiple equilibria exist.

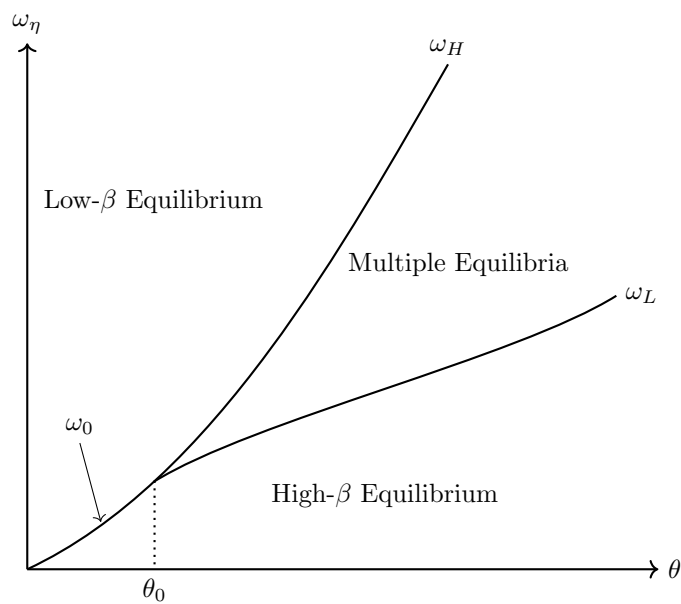


Figure 3: Equilibrium Characterization

Note: This figure illustrates the equilibrium states using the flow-performance sensitivity ( $\theta$ ) and fund opacity ( $\omega_\eta$ ). When  $\theta \leq \theta_0$ , the cutoffs of opacity converge to  $\omega_H = \omega_L = \omega_0$ .

hiding and the showing-off motive tends to balance and generates multiple equilibria when the fund is moderately opaque and fund flows are sensitive to the expected investment performance. Otherwise, one of the conflicting motives dominates the other, making either the Kyle-like equilibrium or the flow-driven equilibrium the unique equilibrium.

*Equilibrium comparison.* Comparing the two stable equilibria, does the manager perform better in one of them? What are the implications for market liquidity, asset prices, and learning about asset-selection skills?

**Corollary 3.2** (High- $\beta$  vs. low- $\beta$  equilibrium). *In the high- $\beta$  equilibrium, relative to the low- $\beta$  equilibrium, the following results hold.*

- (i) *The market is more liquid, that is,  $\lambda$  is lower;*
- (ii) *The manager's expected investment performance given the realized skill,  $E[(\delta - p)x|s] = \frac{\beta\omega_u s^2}{\beta^2\omega_s + \omega_u}$ , is lower;*
- (iii) *The price informativeness, defined by  $\tau \equiv \frac{1}{\text{Var}[\delta|p]} = \frac{\beta^2\omega_s + \omega_u}{\omega_u\omega_s}$ , is higher;*
- (iv) *The price volatility,  $\text{Var}[p] = \frac{\beta^2\omega_s^2}{\beta^2\omega_s + \omega_u}$ , is higher; and*
- (v) *Investors' estimate of the manager's skill is more precise, that is,  $\hat{\omega}_s = \text{Var}[s|f] = \frac{\omega_s\omega_\eta}{\beta^2\omega_s + \omega_\eta}$  is lower.*

Statement (i) of Corollary 3.2 holds because price impact  $\lambda$  is decreasing in  $\beta$  in the equilibrium. Namely, the information is exaggerated in that the order size is too large relative to the true  $|s|$  due to the showing-off motive. Thus, the market maker attempts to partially “undo” the manager’s information revelation by incorporating less of it into the price, resulting in a weaker price impact. Since the high- $\beta$  equilibrium emerges only if the fund is relatively transparent, the statement also implies that a higher fund transparency leads to higher market liquidity.

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<sup>15</sup> $\lambda$  represents the price’s reaction to the order flow  $x + u$ , rather than its reaction to the manager’s private information  $s$ . The price impounds  $s$  with coefficient  $\beta\lambda$ , which is monotonically increasing in  $\beta$ .

The intuition for statement (ii) is closely related to that for statement (i). In the high- $\beta$  equilibrium, the manager earns a lower trading profit due to a smaller informational advantage over the market maker. This result may appear inconsistent with statement (i) because, *ceteris paribus*, a lower price impact is typically associated with a higher trading profit. However, the manager's order size  $|x|$  in the high- $\beta$  equilibrium is so large that the *overall* price reaction (i.e.,  $\lambda$  multiplied by  $|x|$ ) is larger than that in the low- $\beta$  equilibrium. Consequently, the manager moves the price excessively and earns a lower trading profit in this equilibrium. Due to her aggressive trading, more information about  $s$ , which is informative about  $\delta$ , is incorporated into the price in the high- $\beta$  equilibrium, supporting statement (iii).

Statement (iv) follows, again, from the manager's aggressive trading in the high- $\beta$  equilibrium. Specifically, she places a large buy (sell) order when  $s > 0$  ( $s < 0$ ), even if  $|s|$  is small. This causes a large upward (downward) pressure on  $p$ , leading to greater price variability. This result aligns with the insight of Guerrieri and Kondor (2012) that reputation concerns amplify price volatility, although their mechanism differs from ours.<sup>16</sup>

In the high- $\beta$  equilibrium, the manager reveals more information about  $s$ , thereby improving the precision of investors' estimation, as statement (v) demonstrates. This makes fund flows more responsive to skill, as investors can assess it more accurately.

## 3.7 Implications

This section discusses economic implications drawn from the equilibrium analyses under opacity.

### 3.7.1 Financial Fragility

The self-fulfilling nature of multiple equilibria suggests that financial markets may become vulnerable to non-fundamental factors. A shift in investor beliefs

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<sup>16</sup>Guerrieri and Kondor (2012) argue that bond price volatility increases because fund managers demand a premium to offset the risk of reputation damage in the event of default.

alone, without any structural changes, could cause the market to transition from the standard Kyle-like equilibrium to the flow-driven equilibrium, characterized by high price volatility and poor trading performance. We interpret this phenomenon as a form of *fragility* in financial markets.

Our model suggests that fragility may occur when flow-performance sensitivity ( $\theta$ ) is sufficiently high. This result is consistent with empirical studies documenting that strong flow-performance sensitivity can amplify trading pressure and contribute to market fragility by inducing managers to adjust trading behavior in response to investor flows rather than fundamentals (e.g., Coval and Stafford, 2007; Goldstein, Jiang, and Ng, 2017). While this literature typically attributes fragility to flow dynamics without explicitly focusing on the size of the investor base, our results suggest that a larger clientele ( $n$ ) mechanically increases  $\theta$  and thereby sows the seeds of market fragility.

A high flow-performance sensitivity can also arise when the fund management fee ( $\phi$ ) is relatively low. This surprising result goes counter to the recent debate on performance-based compensations for fund managers and advisory contracts, where payment structures that strongly reward performance are often criticized for encouraging excessive risk-taking. In our model, such performance-based compensation can actually stabilize the market by eliminating the flow-driven equilibrium and guiding the economy toward a unique low- $\beta$  equilibrium. This is consistent with the empirical finding by Dass, Massa, and Patgiri (2008) that high-powered advisory contracts do not induce investment in bubbly stocks but instead reduce such investments, highlighting the stabilizing effect of incentive compensations on financial markets.

### 3.7.2 Opacity and Trading Style

Furthermore, Proposition 3.1 yields novel implications connecting the fund opacity and its trading styles.

**Corollary 3.3.** *Both in the high- and low- $\beta$  equilibria, the trading intensity  $\beta$  is monotonically decreasing in the fund opacity  $\omega_\eta$ .*

Figure 4 illustrates the result in Corollary 3.3: the left panel represents



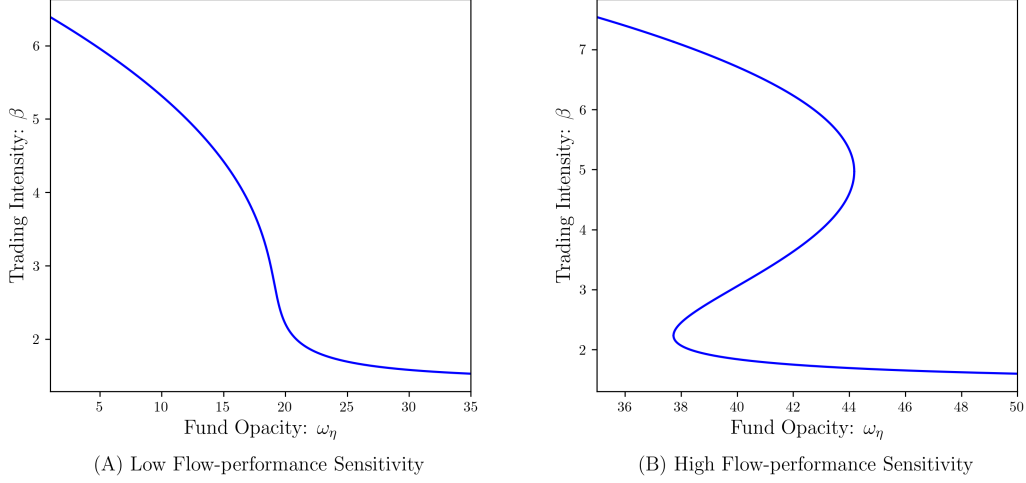


Figure 4: Fund Opacity and Trading Intensity

Note: These figures are illustrated by setting  $\omega_s = 1.0, \omega_u = 2.0, \theta = 10 (\phi = 0.08)$  for the left panel and  $\theta = 27 (\phi = 0.02)$  for the right panel.

the case with a unique equilibrium by setting a weak flow-performance sensitivity, while the right panel involves multiple equilibria due to a strong flow-performance sensitivity. First, as both panels illustrate, opaque funds tend to adopt low-intensity strategies (i.e., the low- $\beta$  equilibrium) that minimize market impact, prioritizing profits over fund flows. In contrast, transparent funds engage in high-intensity strategies with flashy risk taking (i.e., the high- $\beta$  equilibrium), seeking reputation among investors and large fund flows. This relationship arises because the manager's reputation becomes insensitive to her trading behavior when the fund information is opaque and difficult to interpret for fund investors. The model predicts that these tendencies are more pronounced when the flow-performance sensitivity is weak, as the equilibrium is unique regardless of fund opacity (panel [a] of Figure 4).

These results are consistent with common observations in the real market. For example, quantitative hedge funds are often extremely opaque and highly profit-driven, focusing on consistent performance, with benefits from attracting fund flows being regarded as secondary to profitability (Schwager, 2012). On the other hand, some transparent funds, such as certain thematic ETFs or

crypto-focused funds, engage in bold, attention-grabbing strategies to attract investor flows. These funds often emphasize visibility and narrative-driven positioning, which can lead to volatile returns and greater risk taking.

The second implication is drawn from the model’s multiple equilibria, as panel (b) in Figure 4 illustrates. Namely, the relation between funds’ activities ( $\beta$ ) and their transparency ( $\omega_\eta$ ) may become blurry when the opacity is at an intermediate level. Notably, this indeterminacy may lead to financial fragility. Depending on market beliefs, relatively opaque funds may shift between profit-seeking and flow-driven strategies, creating non-fundamental fluctuations. Our result suggests that even relatively opaque funds may follow the crowd due to market beliefs, hopping between safe and risky strategies.

A notable observation in line with this result is momentum crashes (Daniel and Moskowitz, 2016), where funds engaging in risky and visible momentum trades often switch to safer strategies, generating sharp reversals and amplifying price declines. More generally, the rapid shifts between risk-taking and solid performance-based behavior have contributed to market fragility and volatility, as documented in various studies on economic crises, such as Brunnermeier (2009) on the Global Financial Crisis in 2007. Importantly, large fundamental shifts that support such transitions are rarely identified (Gennotte and Leland, 1999). Our model does not require such fundamental shocks, as a change in beliefs alone can trigger a shift.

### 3.7.3 Compensation in Fund Management Industry

Our model provides insights into how skills are reflected in compensation in the fund management industry. As shown in statement (v) of Corollary 3.2, fund investors learn  $s$  more precisely in the high- $\beta$  equilibrium than in the low- $\beta$  equilibrium, as the manager demonstrates her skill by trading more intensively. Consequently, in the high- $\beta$  equilibrium, the manager’s expected compensation,  $E[\phi y|s]$ , become more sensitive to her realized skill. As an illustration, Figure 5 plots the manager’s expected compensation in relation to her skill in the low- and high- $\beta$  equilibria. In the high- $\beta$  equilibrium, investors’ accurate evaluations result in greater pay inequality: low-skilled managers

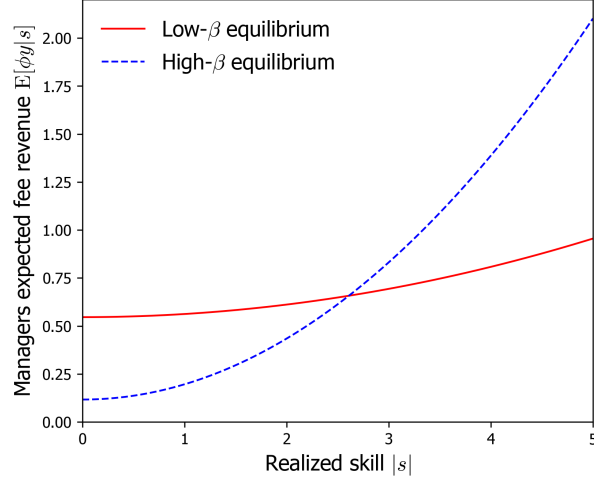


Figure 5: Skill versus Expected Wage

Note: The figure plots the manager's expected fee income at the low- $\beta$  equilibrium (solid line) and high- $\beta$  equilibrium (dashed line) for different levels of skill,  $|s|$ . The parameter values used in the figures are  $\omega_s = 1$ ,  $\omega_u = 2$ ,  $\omega_\eta = 38$ , and  $\theta = 27$ .

would receive very low compensation, while high-skilled ones are very highly paid. This result corroborates empirical observations, such as those provided by Philippon and Reshef (2012) and Böhm, Metzger, and Strömberg (2018), highlighting the comparatively large wage inequality in the fund management industry. Célérier and Vallée (2019) attribute its source to the scalability of skill, while inequality in our model arises from managers' incentives to attract fund flows through establishing high reputation.

### 3.7.4 Skill and Performance Persistence

Do high-skilled managers consistently perform well? Not always: skilled managers may not necessarily outperform less skilled ones. Consider a highly skilled manager and a relatively less skilled one. Since trading profits are increasing in managers' skill  $|s|$ , the former is expected to achieve higher trading profits than the latter *conditional on* both managers trading at the same intensity  $\beta$ . However, if their investors believe, for whatever reason, that the skilled manager invest more aggressively than the less skilled one, and managers con-

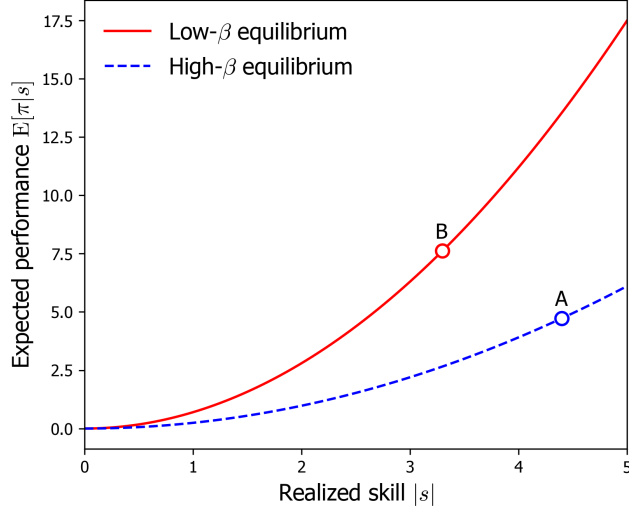


Figure 6: Skill versus Expected Trading Profit

Note: The figure plots the expected trading profit  $E[\pi]$  at the low- $\beta$  equilibrium (solid line) and the high- $\beta$  equilibrium (dashed line) for different levels of skill  $|s|$ . The parameter values used in the figure are  $\omega_s = 1$ ,  $\omega_u = 2$ ,  $\omega_\eta = 38$ , and  $\phi = 0.02$ .

form to such beliefs, a high skill level does not translate into performance. This is because the skilled manager, trading at the flow-driven equilibrium, reveals too much private information to the market maker. Hence, compared to the less skilled manager, who plays the Kyle-like equilibrium, the skilled manager may earn lower expected profits. Figure 6 illustrates such a situation.<sup>17</sup> It plots the expected trading profits (performance) at high- and low- $\beta$  equilibria with different levels of skill, suggesting that a low-skilled manager in the low- $\beta$  equilibrium (point B) outperforms a high-skill manager in the high- $\beta$  equilibrium (point A).

The empirical literature (e.g., Jensen, 1967; Busse, Goyal, and Wahal, 2010) finds that asset managers do not outperform passive benchmarks on average and, in cross-section, those who outperform cannot do so consistently. While the literature attributes the lack of persistent performance to the lack of skill, our result proposes an alternative explanation. Several studies reconcile

<sup>17</sup>The same argument holds for the comparison of the unconditional expected profit,  $E[\pi]$ , using skill variance  $\omega_s$  as an *ex-ante* measure of the skill level.

the existence of skill with the lack of persistent performance. [Berk and Green \(2004\)](#), for example, argue that skill-based alphas eventually disappear due to competition between investors and decreasing returns to scale at the fund level.<sup>18</sup> Our model contributes to the understanding of this matter by showing that the lack of interplay between manager performance and skill could be the result of self-fulfilling multiple equilibria.

## 4 Strategic Obfuscation

This section studies strategic obfuscation by the fund manager, who controls  $\omega_\eta$ . Our results shed light on the equilibrium relationship among fund opacity, manager’s skill, and trading style.

### 4.1 Setup

*Obfuscating fund information.* Prior to investors’ capital allocation and the manager’s investment decision, the fund manager chooses the fund opacity  $\omega_\eta$ , anticipating the equilibrium reactions of investors and the market maker specified in Section 3. We interpret this behavior as strategic obfuscation by the fund manager. For example, in many mutual funds, managers provide information about their strategies to investors in accordance with regulatory requirements. However, they strategically design the way this information is presented and convey a deliberately uninformative description even when the underlying holdings are simple. We assume that such practices take costs: the manager pays the obfuscation cost,  $C(\omega_\eta)$  with  $C'(\cdot) > 0$  and  $C(0) = 0$ , to increase the degree of opacity of fund information. It could be thought of as pecuniary costs arising from investments in communication and marketing resources to design abstract narratives and vague labels that replace direct descriptions of risky portfolios. It may also reflect non-pecuniary effort that

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<sup>18</sup>[Pastor, Stambaugh, and Taylor \(2015\)](#) document increases in skill in finance and attribute the lack of corresponding increases in performance to decreasing returns to scale at the industry level.

managers incur to carefully structuring disclosure documents to maintain opacity while following regulatory compliance. In what follows, we assume that the marginal cost of opacity  $C'(\cdot)$  is sufficiently large to ensure the second-order condition of the manager's problem defined below.

*Sunspot shock.* To formalize the manager's optimization problem in the rational expectations framework, we introduce an equilibrium-selection device in cases of multiple equilibria. In particular, following the most common approach, we assume that market participants condition their actions on the realization of an exogenous *sunspot* shock, characterized by the binary random variable  $z \in \{0, 1\}$ .<sup>19</sup> Namely, whenever parameters support multiple equilibria, market participants play the high- $\beta$  equilibrium if a spot appears on the sun ( $z = 1$ ), whereas the low- $\beta$  equilibrium is realized if no spots are observed ( $z = 0$ ). Accordingly, the high- and low- $\beta$  equilibria are realized with probability  $\rho_H \equiv \Pr(z = 1)$  and  $\rho_L = 1 - \rho_H$ , respectively. The shock is realized after the fund manager sets fund opacity but prior to the fund-raising stage.

*Manager's expected utility.* Incorporating the cost, the manager controls  $\omega_\eta$  to maximize her *ex-ante* expected utility derived from the fee income  $E[\phi y]$ , net of the obfuscation cost  $C$ . As the fund flow is given by equation (3.7), her objective function is specified as follows.<sup>20</sup>

$$U = \phi E[\pi + M] - C(\omega_\eta) = \phi(1 + \theta)E[\pi] - C(\omega_\eta), \quad (4.1)$$

where the expectation operator incorporates the sunspot shock together with all other random variables introduced so far.

*Equilibrium.* In this extended model, the definition of equilibrium augments Definition 1 of the baseline model by additionally requiring that  $\omega_\eta$  maximizes

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<sup>19</sup>The sunspot equilibrium is proposed by Diamond and Dybvig (1983) as an equilibrium selection device in the context of bank runs and formalized in the equilibrium analyses by the subsequent studies, such as Cooper and Ross (1998).

<sup>20</sup>The law of iterated expectation implies  $E[M] = \theta E[E[\pi|f]] = \theta E[\pi]$ .

the manager’s objective function  $U$  in equation (4.1).

## 4.2 Opacity as Commitment Device

Before solving problem (4.1), consider the manager’s obfuscation incentive. Given the trading intensity  $\beta$ , the unconditional expectation of fund performance in (4.1) is computed as

$$E[\pi] = \frac{\beta\omega_u}{\beta^2\omega_u + \omega_s}\omega_s. \quad (4.2)$$

As in the standard Kyle (1985) framework, the first fraction represents the profit margin that each unit of informational advantage would earn, and the second term  $\omega_s$  represents the expected size of this advantage.

Fund opacity  $\omega_\eta$  does not directly show up in equation (4.2), but it influences the expected performance through the equilibrium trading intensity  $\beta$ , as Corollary 3.3 demonstrates.

**Corollary 4.1.** *Both in the low- and high- $\beta$  equilibria, the expected fund performance  $E[\pi]$  is monotonically increasing in fund opacity  $\omega_\eta$ .*

Figure 7 plots  $E[\pi]$  in relation to  $\omega_\eta$ , which traces the reactions of  $\beta$  in Figure 4. In our model, holding other parameters fixed,  $E[\pi]$  declines with the trading intensity  $\beta$ . This arises from the presence of the showing off motive. Although unconditional expected performance is maximized at the Kyle benchmark, which is driven solely by the hiding motive, the manager in our model trades more aggressively in order to attract fund flows. It reveals too much private information to the market maker and lowers the fund performance, while the manager is willing to deviate from the profit-maximizing strategy because the fund flow compensates for this loss. The opacity, conversely, diminishes the showing-off motive and induces the manager to take more profit-driven strategies, bringing  $\beta$  closer to the profit-maximizing level. As a consequence, opacity helps improve the expected fund performance and, according to the manager’s utility (4.1), this effect encourages the manager to obfuscate fund information.

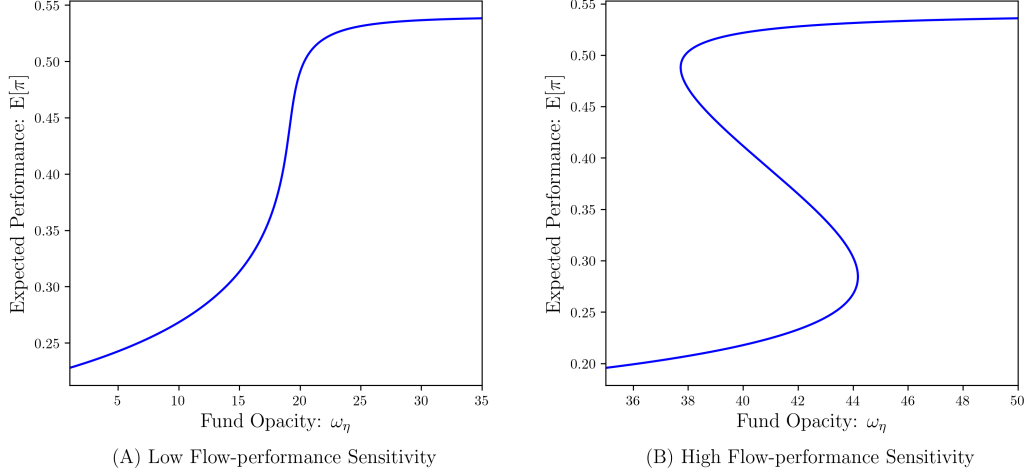


Figure 7: Fund Opacity and Expected Performance

Note: The figure plots the unconditional expected performance of the fund  $E[\pi]$  against fund opacity  $\omega_\eta$  with different values of the flow-performance sensitivity:  $\theta = 10$  (the left panel) and  $\theta = 27$  (the right panel). The value of other parameters are  $\omega_s = 1.5$  and  $\omega_u = 3$ .

Interestingly, investors' *ex-ante* expected utility is also increasing in fund opacity. According to equation (3.6), each investor expects to obtain

$$U_i \equiv \frac{1 - \phi(1 + \theta)}{n} E[\pi], \quad (4.3)$$

so that her expected utility is proportional to the fund's unconditional expected performance. Thus, Corollary 4.1 implies that even investors prefer opaque fund from the *ex-ante* perspective. This result may appear paradoxical in light of the showing-off motive: why does the manager trade aggressively and deviate from the performance-maximizing strategy, despite knowing that such behavior lowers performance and reduces both her own and investors' expected utility? This tension arises from a crucial information timing gap. When she decides on the trading strategy  $x$ , the manager cannot alter the market's beliefs about her strategy ( $\beta$  and  $\lambda$  in equations [3.8]–[3.11]), meaning that she optimizes over  $x$  by taking the investors' belief-updating rule as given. Since she can influence investor perceptions only through the content of fund information  $f$ , she is inclined to inflate her trading volume  $|x|$  to appear



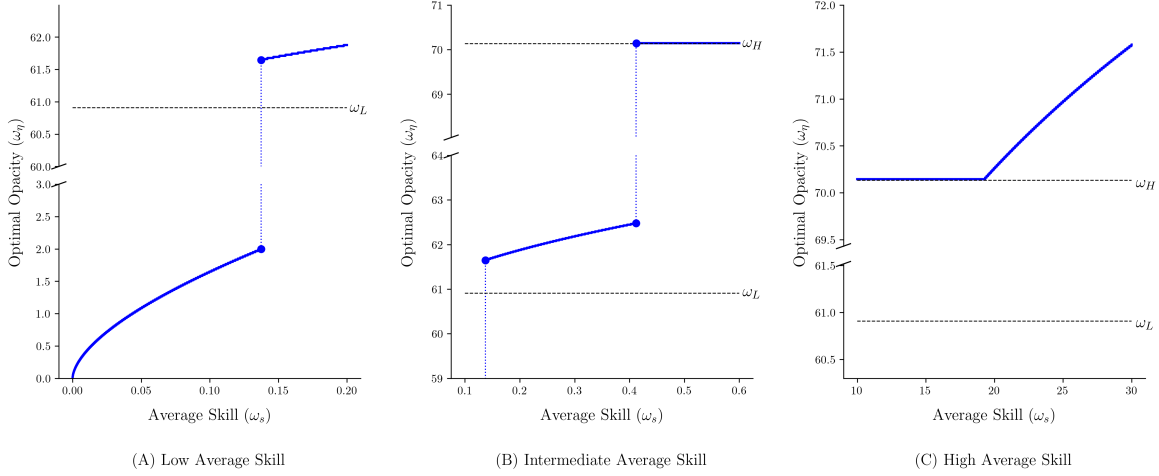


Figure 8: Average Skill and Optimal Obfuscation

Note: The figure plots the optimal degree of fund opacity  $\omega_\eta$  against the manager's expected (average) skill  $\omega_s$ , focusing on three regions that correspond to the equilibrium types. We use the parameter values  $\phi = 0.03, n = 5, \omega_u = 5, \rho_H = 0.10$ , and the obfuscation cost is  $C = c\omega_\eta^\gamma$  with  $c = 10^{-4}$  and  $\gamma = 1.85$ .

more skilled, as captured by the showing-off motive. Conversely, at the *ex-ante* stage (i.e., before she sends the fund information to investors), the manager recognizes that this future showing-off motive will lead to flashy but inefficient risk-taking that erodes unconditional performance and the expected fund flow. Consequently, she strategically chooses fund opacity as a commitment device to “tie her own hands,” thereby dampening the investors' future responsiveness to  $f$  and forcing her future self to prioritize profit-driven trading over flow-driven trading.

### 4.3 Equilibrium Opacity

1. Present the numerical results in Figure 8. The main takeaway here is the optimal degree of obfuscation is globally increasing in the expected skill. This is because obfuscation benefits the manager by mitigating the showing-off motive and guiding her to take more performance-based strategies. It prevents revelation of private information to the market

maker and improves the profit margin of the informational advantage. This positive effect of obfuscation on the manager’s utility is larger when her expected skill is higher, shaping the optimal obfuscation an increasing function of the expected skill.

2. The optimal opacity may involve jumps and stay constant. This is due to multiple equilibria: when  $\omega_s$  switches from the region with a unique high- $\beta$  equilibrium to that with multiple equilibria, the manager’s expected utility starts incorporating the low- $\beta$  equilibrium happening with probability  $\rho_L$ . From Corollary XXX, the manager’s expected utility experiences a positive jump
3. Implications: relationship between managers skill, opacity, and their trading style, including when financial fragility matters (or who contributes to fragility).
  - (a) For example, the jump of opacity when the economy switches from the multiple equilibria region to the unique low- $\beta$  equilibrium region has testable implications for the fund opacity, skill, trading styles, and expected performance. Our results suggest that distribution of managers’ obfuscation and skills would exhibit a discontinuity or clustering around the threshold  $\omega_H$ .

#### 4.3.1 Skill, Opacity, and Fragility

## 5 Conclusion

This paper studies a model à la [Kyle \(1985\)](#) where traders are privately informed about their skill, i.e., the trader-specific ability to identify potentially mispriced assets. The traders care not only about trading profits but also the recruiter’s perception of their skill. The model has two stable self-fulfilling equilibria, which we interpret as fragility in financial markets. One of the equilibria is similar to that of [Kyle \(1985\)](#) where reputation is irrelevant. In the other, “reputation-driven” equilibrium, the traders trade more aggressively for fear of being seen as unskilled by the recruiter, causing higher price volatility,

higher market liquidity, and lower trading profits. Our model predicts that fragility may arise when poaching is (moderately) active in the finance industry and when there are high demands for financial skill. The model yields implications connecting traders' skills, their trading styles, and financial fragility. Moreover, consistent with empirical observations, our model predicts large pay inequality in the finance industry and reconciles skill in active management and the lack of performance persistence.

As a direction for future research, it would be interesting to analyze various types of wage structures in financial markets, ranging from back-end to front-end compensations. The implications of the forward-looking trading motives introduced in our model can be applied to investigate which types of wage structures and employment systems can mitigate financial market fragility while also improving the welfare of market participants.

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# Online Appendix

## A Proof of Lemma ??

From (3.1) and (3.2),  $p$  is observationally equivalent to  $\frac{p-\hat{\delta}}{\lambda} = \beta s + u \equiv z_p$ , which is the sum of two normally distributed variables from the market maker's perspective. Since  $\delta \sim N(\hat{\delta} + s, \omega_\delta)$  is equivalently presented as  $\delta = \hat{\delta} + s + \epsilon$  with  $\epsilon \sim N(0, \omega_\delta)$ ,  $\delta$  is observationally equivalent to  $\delta - \hat{\delta} = s + \epsilon \equiv z_\delta$ , which is also the sum of normal variables. Thus, given  $Z = [z_p, z_\delta]^T$ , the market maker's posterior mean of  $s$  is

$$\mathbb{E}[s|Z] = \underbrace{\mathbb{E}[s]}_{=0} + \Sigma_{sZ} \Sigma_Z^{-1} (Z - \underbrace{\mathbb{E}[Z]}_{=[0 \ 0]^T}), \quad (\text{A.1})$$

where

$$\Sigma_{sZ} = \begin{bmatrix} \beta\omega_s & \omega_s \end{bmatrix} \text{ and } \Sigma_Z = \begin{bmatrix} \beta^2\omega_s + \omega_u & \beta\omega_s \\ \beta\omega_s & \omega_s + \omega_\delta \end{bmatrix}. \quad (\text{A.2})$$

Since

$$\begin{aligned} \det[\Sigma_Z] &= (\beta^2\omega_s + \omega_u)(\omega_s + \omega_\delta) - \beta^2\omega_s^2 \\ &= \omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta, \end{aligned} \quad (\text{A.3})$$

it holds that

$$\begin{aligned} \mathbb{E}[s|Z] &= \begin{bmatrix} \beta\omega_s & \omega_s \end{bmatrix} \frac{1}{\det[\Sigma_Z]} \begin{bmatrix} \omega_s + \omega_\delta & -\beta\omega_s \\ -\beta\omega_s & \beta^2\omega_s + \omega_u \end{bmatrix} \begin{bmatrix} z_p \\ z_\delta \end{bmatrix} \\ &= \frac{\omega_s}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} (\beta\omega_\delta z_p + \omega_u z_\delta) \\ &= \frac{\omega_s(\beta\omega_\delta + \omega_u)}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \left( \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u} z_p + \left(1 - \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u}\right) z_\delta \right) \\ &= \xi (\gamma z_p + (1 - \gamma) z_\delta), \end{aligned} \quad (\text{A.4})$$



where

$$\xi \equiv \frac{\omega_s(\beta\omega_\delta + \omega_u)}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \quad (\text{A.5})$$

and

$$\gamma \equiv \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u}. \quad (\text{A.6})$$

Re-writing  $z_p$  and  $z_\delta$  in (A.4) in terms of  $p$  and  $\delta$ , we obtain the required result.  $\square$

## B Proof of Lemma ??

Let us present  $\delta \sim N(\hat{\delta} + s, \omega_\delta)$  equivalently as  $\delta = \hat{\delta} + s + \epsilon$  with  $\epsilon \sim N(0, \omega_\delta)$ . Then, squaring (??), we have

$$\begin{aligned} \hat{s}^2 &= \xi^2 \left( \gamma^2 x^2 + 2\gamma x \left( \gamma u + (1 - \gamma)(\delta - \hat{\delta}) \right) + \left( \gamma u + (1 - \gamma)(\delta - \hat{\delta}) \right)^2 \right) \\ &= \xi^2 \left( \begin{array}{c} \gamma^2 x^2 + 2\gamma^2 x u + 2\gamma(1 - \gamma)(s + \epsilon)x \\ + \gamma^2 u^2 + 2\gamma(1 - \gamma)(s + \epsilon)u + (1 - \gamma)^2(s + \epsilon)^2 \end{array} \right). \end{aligned} \quad (\text{B.1})$$

Taking the expectation of (B.1) from the trader's perspective,

$$\text{E}[\hat{s}^2|s] = \xi^2 \left( \gamma^2 x^2 + 2\gamma(1 - \gamma)sx + A \right), \quad (\text{B.2})$$

where

$$A \equiv \gamma^2 \omega_u + (1 - \gamma)^2(s^2 + \omega_\delta) \quad (\text{B.3})$$

is the term that the trader is unable to influence by her choice of  $x$ . We can omit  $A$  in the maximization problem below. Using (B.2), the trader's problem (??) is rewritten as

$$\begin{aligned} &\max_x \text{E}[(\delta - p)x|s] + \theta \text{E}[\hat{s}^2|s] \\ \Rightarrow &\max_x \text{E} \left[ \left( \hat{\delta} + s + \epsilon - \left( \hat{\delta} + \lambda(x + u) \right) \right) x \middle| s \right] + \theta \text{E}[\hat{s}^2|s] \\ \Rightarrow &\max_x sx - \lambda x^2 + \theta \xi^2 \left( \gamma^2 x^2 + 2\gamma(1 - \gamma)sx \right). \quad \square \end{aligned} \quad (\text{B.4})$$

## C Proof of Lemma ??

At the beginning of  $t = 1$ , the trader does not have reputation concerns anymore. Her problem is just to choose  $x_1$  to maximize the expected trading profit, that is,

$$\max_{x_1 \in (-\infty, \infty)} \mathbb{E}[(\delta_1 - p_1)x_1 | s]. \quad (\text{C.1})$$

We conjecture and later verify that the trader's equilibrium order in  $t = 1$  is

$$x_1 = \beta_1 s, \quad (\text{C.2})$$

where  $\beta_1 > 0$  is a constant to be determined. Given (C.2), the market maker sets the price

$$\begin{aligned} p_1 &= \mathbb{E}[\delta_1 | x_1 + u_1] \\ &= \mathbb{E}[\mathbb{E}[\delta_1 | x_1 + u_1, s] | x_1 + u_1] \\ &= \mathbb{E}[\hat{\delta}_1 + s | x_1 + u_1] \\ &= \hat{\delta}_1 + \frac{\text{Cov}[s, x_1 + u_1]}{\text{Var}[x_1 + u_1]} (x_1 + u_1 - \mathbb{E}[x_1 + u_1]) \\ &= \hat{\delta}_1 + \frac{\text{Cov}[s, \beta_1 s + u_1]}{\text{Var}[\beta_1 s + u_1]} (x_1 + u_1 - \mathbb{E}[\beta_1 s + u_1]) \\ &= \hat{\delta}_1 + \lambda_1 (x_1 + u_1), \end{aligned} \quad (\text{C.3})$$

where

$$\lambda_1 \equiv \frac{\beta_1 \omega_s}{\beta_1^2 \omega_s + \omega_{u,1}}. \quad (\text{C.4})$$

Given (C.3), the trader's problem (C.1) is rewritten as

$$\max_{x_1} \mathbb{E} \left[ \left( \delta_1 - \left( \hat{\delta}_1 + \lambda_1 (x_1 + u_1) \right) \right) x_1 \middle| s \right] \Rightarrow \max_{x_1} s x_1 - \lambda_1 x_1^2. \quad (\text{C.5})$$

The FOC for  $x_1$  is  $s - 2\lambda_1 x_1 = 0$ , which determines the trader's optimal action:

$$x_1 = \frac{1}{2\lambda_1} s. \quad (\text{C.6})$$

For the other agents' belief (C.2) to be consistent with (C.6), we need

$$\beta_1 = \frac{1}{2\lambda_1} \iff \beta_1 = \sqrt{\frac{\omega_{u,1}}{\omega_s}}, \quad (\text{C.7})$$

which implies

$$\lambda_1 = \frac{1}{2} \sqrt{\frac{\omega_s}{\omega_{u,1}}}. \quad \square \quad (\text{C.8})$$

## D Proof of Lemma ??

Let us present  $\delta_1 \sim N(\hat{\delta}_1 + s, \omega_\delta)$  equivalently as  $\delta_1 = \hat{\delta}_1 + s + \epsilon_1$  with  $\epsilon_1 \sim N(0, \omega_\delta)$ . Then the trader's trading profit in  $t = 1$  is

$$\begin{aligned} \pi_1 &= (\delta_1 - p_1)x_1 \\ &= \left( \hat{\delta}_1 + s + \epsilon_1 - \left( \hat{\delta}_1 + \lambda_1 (\beta_1 s + u_1) \right) \right) \beta_1 s \\ &= \left( \hat{\delta}_1 + s + \epsilon_1 - \left( \hat{\delta}_1 + \frac{1}{2} \sqrt{\frac{\omega_s}{\omega_{u,1}}} \left( \sqrt{\frac{\omega_{u,1}}{\omega_s}} s + u_1 \right) \right) \right) \sqrt{\frac{\omega_{u,1}}{\omega_s}} s \\ &= \frac{1}{2} \sqrt{\frac{\omega_{u,1}}{\omega_s}} s^2 + \left( \sqrt{\frac{\omega_{u,1}}{\omega_s}} \epsilon_1 - \frac{u_1}{2} \right) s. \end{aligned} \quad (\text{D.1})$$

Taking the expectation of (D.1) from the recruiter's perspective, we obtain the trader's salary:

$$\begin{aligned} w &= \mathbb{E}[\pi_1 | p_0, \delta_0] \\ &= \frac{1}{2} \sqrt{\frac{\omega_{u,1}}{\omega_s}} \mathbb{E}[s^2 | p_0, \delta_0] \\ &= \frac{1}{2} \sqrt{\frac{\omega_{u,1}}{\omega_s}} (\mathbb{E}[s | p_0, \delta_0]^2 + \text{Var}[s | p_0, \delta_0]) \\ &= \frac{1}{2} \sqrt{\frac{\omega_{u,1}}{\omega_s}} (\hat{s}^2 + \hat{\omega}_s), \end{aligned} \quad (\text{D.2})$$

where  $\hat{\omega}_s$  is the posterior variance of  $s$ , which is a constant.  $\square$

## E Proof of Propositions 3.1 and ??

In the following analyses, denote

$$a = \frac{\omega_{u,0}}{\omega_s}, \quad b = \frac{\omega_{u,0}}{\omega_\delta}, \quad (\text{E.1})$$

so that we rewrite equation (3.15) as

$$\beta = F(\beta) \equiv \frac{1 + 2\theta \left( \frac{1}{a+b+\beta^2} \right)^2 b\beta}{2 \left( \frac{\beta}{\beta^2+a} - \theta \left( \frac{\beta}{a+b+\beta^2} \right)^2 \right)}. \quad (\text{E.2})$$

By rearranging (E.2), the equilibrium  $\beta$  solves the following condition.

$$2\theta\beta = J(\beta^2), \quad (\text{E.3})$$

where, by denoting  $\beta^2 = y$ ,

$$J(y) = \frac{(y-a)(y+b+a)^2}{(y+a)(y+b)}. \quad (\text{E.4})$$

The first-order derivative of the RHS of (E.3) with respect to  $\beta$  is

$$K^{(1)}(\beta) \equiv 2\beta J^{(1)}(\beta^2), \quad (\text{E.5})$$

with

$$J^{(1)}(y) \equiv \frac{N(y)}{(y+a)^2 (y+b)^2}, \quad (\text{E.6})$$

where the numerator is

$$N(y) = y^4 + 2(a+b)y^3 + n_2y^2 + n_1y + n_0$$

with coefficients being

$$\begin{aligned} n_2 &= 2a^2 + 6ab + b^2, \\ n_1 &= 2a [3b^2 + 3ab + a^2], \end{aligned}$$

$$n_0 = a (a^2 + ab + 2b^2) (a + b).$$

From (E.4) and (E.6),  $y = a$  is the unique solution to  $J(y) = 0$  in  $y > 0$ , and  $J(y)$  is increasing in  $y$  for  $y \geq a$ . Hence, there is no equilibrium in  $y < a$ , and we focus on  $y \geq a$ .

The second-order derivative of the RHS of (E.3) with respect to  $\beta$  is expressed as

$$K^{(2)}(y) \equiv 2 (J^{(1)}(y) + 2yJ^{(2)}(y)), \quad (\text{E.7})$$

where  $J^{(2)}$  denotes the second-order derivative of  $J$  with respect to  $y$ .  $K^{(1)}$  and  $K^{(2)}$  have the following properties:

**Lemma E.1.** (i)  $K^{(1)}(\beta)$  takes a U-shaped curve in relation to  $\beta$ .  
(ii)  $K^{(2)}(a) < 0$  and  $K^{(2)}(\infty) > 0$ . There exists a unique solution to  $K^{(2)}(y) = 0$ .

*Proof.* Note that point (i) directly follows from point (ii). By denoting that  $N^{(1)}(y) = \frac{dN(y)}{dy}$ , it holds that

$$J^{(1)}(y) + 2yJ^{(2)}(y) = J^{(1)}(y) \frac{-7y^2 - 3(a+b)y + ab}{(y+a)(y+b)} + 2y \frac{(y^2 + (a+b)y + ab) N^{(1)}(y)}{(y+a)^3 (y+b)^3}. \quad (\text{E.8})$$

Evaluating at  $y = a$ ,

$$\begin{aligned} J^{(1)}(a) + 2aJ^{(2)}(a) &= -J^{(1)}(a) \frac{5a+b}{a+b} + \frac{N^{(1)}(y)}{2a(a+b)^2} \\ &\propto -2a(a+b)(5a+b)J^{(1)}(a) + N^{(1)}(a) \\ &= -4a^3 - b(4+b)a - b^3 \\ &< 0. \end{aligned}$$

Therefore,  $K^{(2)}(a) < 0$  holds.

The numerator of (E.8) is summarized as

$$T(y) = - \left[ y^4 + 2(a+b)y^3 + n_2y^2 + n_1y + n_0 \right] \left[ 7y^2 + 3(a+b)y - ab \right] \\ + 2y \left( y^2 + (a+b)y + ab \right) \left[ 4y^3 + 6(a+b)y^2 + 2n_2y + n_1 \right].$$

The above argument suggests that  $T(a) < 0$ .

Using Mathematica, we summarize  $T$  into the following polynomial:

$$T(y) = \sum_{k=0}^6 t_k y^k, \quad (\text{E.9})$$

where the coefficients are

$$t_0 = a^2b(a+b) (a^2 + ab + 2b^2) > 0,$$

$$t_1 = -3a (a^4 + (a+b)(a-b)^2b + b^4) < 0,$$

$$t_2 = -3a (3a^3 + 4a^2b + ab^2 + 5b^3) < 0,$$

$$t_3 = -8a^3 - 8a^2b - 9ab^2 + b^3,$$

$$t_4 = 3b(a+b) > 0,$$

$$t_5 = 3(a+b) > 0,$$

and  $t_6 = 1$ . As  $t_6 > 0$ ,  $T(\infty) = \infty > 0$ . Together with  $T(a) < 0$ , the intermediate value theorem implies that  $T(y) = 0$  has at least one solution in  $y > a$ .

Suppose that  $y_{min}$  is one of the solutions to  $T(y) = 0$ . Consider the first-order derivative of  $T$  evaluated at  $y = y_{min}$  and multiplied  $y_{min}$ :

$$\mathcal{T}(y_{min}) = y_{min} \frac{dT(y_{min})}{dy} = \sum_{k=1}^6 k t_k y_{min}^k.$$

Applying  $T(y_{min}) = 0$  and eliminating the terms related to  $t_3$ ,

$$\mathcal{T}(y_{min}) = -3t_0 - 2t_1y_{min} - t_2y_{min}^2 + t_4y_{min}^4 + 2t_5y_{min}^5 + 3t_6y_{min}^6,$$

According to the signs of  $t_k$  and  $y_{min} > a$ , it holds that

$$\begin{aligned}\mathcal{T}(y_{min}) &> -3t_0 - 2t_1a - t_2a^2 + t_4a^4 + 2t_5a^5 + 3t_6a^6 \\ &= 3a^2 (6a^4 + 8a^3b + b^4 + (a^2 - b^2)^2) \\ &> 0.\end{aligned}$$

Therefore, if  $y_{min}$  is a solution to  $T(y) = 0$ , then the slope of  $T(y)$  at this point must be positive. Together with  $T(a) < 0 < T(\infty)$ ,  $y_{min}$  is the unique solution to  $T(y) = 0$ , and  $T(y) > 0 \Leftrightarrow y > y_{min}$ .<sup>21</sup> The argument above implies that  $K^{(1)}(\beta)$  takes a U-shaped curve with respect to  $\beta$ , and the minimum point is given by  $\beta_{min} \equiv \sqrt{y_{min}}$ . Note that  $\tilde{\beta}$  in footnote ?? is given by  $\tilde{\beta} = \beta_{min}$ .  $\square$

The RHS of (E.3) is monotonically increasing in  $\beta$ , and Lemma E.1 attests that it is concave in  $y \leq y_{min}$ , while it becomes convex in  $y > y_{min}$ . In other words, the slope of the curve is the least steep at  $y = y_{min}$ , and it is given by

$$K_{min} = 2\beta_{min}J^{(1)}(y_{min}). \quad (\text{E.10})$$

Note that  $K_{min}$  is represented only by parameters,  $a, b$ , and does not involve  $\theta$ .

Consider the following condition to characterize equilibrium:

$$J(y_{min}) > K_{min}\beta_{min}. \quad (\text{E.11})$$

Then the following result holds:

**Lemma E.2.** *(i) If inequality (E.11) holds, then there exist  $\theta_H$  and  $\theta_L$  with  $\theta_H > \theta_L$ . If, and only if,  $\theta \in \Theta \equiv (\theta_L, \theta_H)$ , then the equilibrium condition (E.3) has three solutions.*

*(ii) If inequality (E.11) is violated, then  $\theta_H = \theta_L$ , and the equilibrium condition (E.3) has a unique solution. The solution is  $\beta > \beta_{min}$  ( $\beta < \beta_{min}$ ) if  $\theta > \theta_H$*

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<sup>21</sup>Suppose that  $T(y) = 0$  has more than two solutions. For any solutions, denoted as  $y_0$  and  $y_1$ , the mean-value theorem implies that there exists at least one  $y_2 \in [y_0, y_1]$  such that  $\frac{dT(y_2)}{dy} = 0$ . This contradicts to the strict positivity or negativity of  $\mathcal{T}$ .

$(\theta < \theta_H)$ .

*Proof.* Consider the plot of the curve,  $J(\beta^2)$ , and the line,  $2\theta\beta$ , against  $\beta$ . With the curve being fixed, consider a line that goes through the origin and is tangent to the curve. The tangent point is characterized by  $\beta$  that satisfies

$$J(\beta^2) = K^{(1)}(\beta^2)\beta,$$

where the LHS represents the y-coordinate of the curve at  $\beta$ , and the RHS represents that of the line that goes through the origin. Note that their slopes must be identical and  $K^{(1)}(\beta^2)$ . Applying (E.5), the above condition is rewritten as

$$J(\beta^2) = 2\beta^2 J^{(1)}(\beta^2).$$

Hence, denoting  $y = \beta^2$ , the tangent point is characterized by the solution to

$$0 = G(y) = 2yJ^{(1)}(y) - J(y).$$

Note that we focus on  $y > \sqrt{a}$ , and thus  $y$  and  $\beta$  have one-to-one mapping.

Function  $G(y)$  has the following first-order derivative:

$$\begin{aligned} G^{(1)}(y) &= 2(yJ^{(2)}(y) + J^{(1)}(y)) - J^{(1)}(y) \\ &= J^{(1)}(y) + 2yJ^{(2)}(y) \\ &= \frac{K^{(2)}(y)}{2}, \end{aligned}$$

where we apply the definition of  $K^{(2)}$  in (E.7). Lemma E.1 has established that  $K^{(2)}(a) < 0$  and  $K^{(2)}(\infty) > 0$ . Also, there is a unique  $y = y_{min}$  such that  $K^{(2)}(y_{min}) = 0$  and  $K^{(2)}(y_{min}) > 0 \Leftrightarrow y > y_{min}$ . Hence,  $G(y)$  takes a U-shaped curve with respect to  $y$  and  $y = y_{min}$  is the minimizer of the curve.

With  $G(y)$  being U-shaped,  $G(y) = 0$  has two solutions if and only if its minimum point dips below the x-axis, namely,

$$2y_{min}J^{(1)}(y_{min}) - J(y_{min}) < 0.$$



Applying the definition of  $K_{min}$  in (E.10), the above inequality is rewritten as condition (E.11).

When condition (E.11) holds and  $G(y) = 0$  has two solutions, we denote them as  $y_0$  and  $y_1(> y_0)$ . The corresponding slopes of these tangent lines are  $K^{(1)}(y_0)$  and  $K^{(1)}(y_1)$ , respectively. Then the original line,  $2\theta\beta$ , and the curve,  $J(\beta^2)$ , have three intersections if and only if

$$\frac{K^{(1)}(y_1)}{2} \equiv \theta_L < \theta < \theta_H \equiv \frac{K^{(1)}(y_0)}{2}. \quad (\text{E.12})$$

This result concludes the proof of statement (i) of the lemma.

Conversely, if condition (E.11) does not hold, no tangent line that goes through the origin is obtained: function  $J(\beta^2)$  becomes the least steep at  $\beta_{min} = \sqrt{y_{min}}$ , and the tangent line is specified by  $K_{min}\beta - K_{min}\beta_{min} + J(\beta_{min}^2)$ , while the intercept of this line is negative, as condition (E.11) is violated. Hence, the line,  $2\theta\beta$ , and the curve,  $J(\beta^2)$  have a unique intersection. This intersection occurs in  $\beta > \beta_{min}$  if, and only if,  $\theta > \theta'_H \equiv \frac{J(\beta_{min}^2)}{2\beta_{min}}$ .

To conclude statement (ii), we must show that  $\theta_H$  and  $\theta_L$  in statement (i) converges to  $\theta'_H$  in statement (ii) at the threshold of these two cases so that we re-define it as  $\theta_H = \theta_L = \theta'_H$ . This result is shown in Lemma E.3 below.  $\square$

Finally, consider the comparative statics with respect to  $\omega_s$ . The following result holds:

**Lemma E.3.** (i)  $J(y_{min}) - K_{min}\beta_{min}$  is increasing in  $\omega_s$ . Hence, there exists a unique  $\tilde{\omega}_s > 0$ , such that, (E.11) holds with equality, and inequality (E.11) holds if and only if  $\omega_s > \tilde{\omega}_s$ .

(ii) When condition (E.11) holds, both  $\theta_H$  and  $\theta_L$  increase as  $\omega_s$  increases. Interval  $\Theta$  expands, as the following inequalities hold:

$$0 < \frac{d\theta_L}{d\omega_s} < \frac{d\theta_H}{d\omega_s}.$$

(iii) At  $\omega_s = \tilde{\omega}_s$ , it holds that  $\theta_H = \theta_L = \theta'_H$ , while at the limit of  $\omega_s \rightarrow 0$ ,  $\theta'_H \rightarrow 0$ .

(iv) There exists a unique threshold for  $\varphi^*$ , such that,  $\varphi \geq \varphi^* (< \varphi^*)$  leads to case 1 (case 2) in Proposition ??.

*Proof of Statements (i) and (ii).* Note that we are interested in the reaction of the following function when  $a = \frac{\omega_{u,0}}{\omega_s}$  changes and, hereafter, we make the dependence of functions on  $a$  explicit:

$$2\theta_k = K^{(1)}(y_j, a) = 2\sqrt{y_j}J^{(1)}(y_j, a),$$

where  $y_j = y_0, y_1$  denote the solutions to  $G(y, a) = 0$ , if any, and  $k = L, H$  are appropriately determined following (E.12). According to the definition of  $G$ , the following partial derivative holds.

$$\frac{\partial G(y, a)}{\partial a} = 2y \frac{\partial J^{(1)}(y, a)}{\partial a} - \frac{\partial J(y, a)}{\partial a}.$$

The first term is summarized by

$$\begin{aligned} \frac{\partial J^{(1)}(y, a)}{\partial a} &\propto (y + a) \frac{\partial N(y, a)}{\partial a} - 2N(y, a) \\ &= (y + a) [2y^3 + n_{2,a}y^2 + n_{1,a}y + n_{0,a}] - [y^4 + 2(a + b)y^3 + n_2y^2 + n_1y + n_0] \\ &= \sum_{k=0}^4 A_k y^k \end{aligned}$$

where  $A_4 = 1$ ,  $A_3 = 4(a + b)$ ,

$$A_2 = 2a^2 + 5b^2 + 2(3ab + a^2) + 2a[3b + 2a],$$

$$A_1 = 2a^2[3b + 2a](a^2 + ab + 2b^2)(2a + b) + a(2a + b)(a + b),$$

$$\begin{aligned} A_0 &= a[(2a + b)(a^2 + ab + 2b^2 + a(a + b)) - (a^2 + ab + 2b^2)(a + b)] \\ &> (a(a + b))^2. \end{aligned}$$

Hence,  $\frac{\partial J^{(1)}(y,a)}{\partial a} > 0$ . The second term is

$$(y+b) \frac{\partial J(y,a)}{\partial a} = -2 \frac{(y+b+a)(y(a+b)+a^2)}{(y+a)^2} < 0. \quad (\text{E.13})$$

Therefore,  $\frac{\partial G}{\partial a} > 0$ . As the previous proof has established that  $G(y,a)$  takes a  $U$ -shaped curve,  $\frac{\partial G}{\partial a} > 0$  implies that its minimum point is monotonically decreasing in  $\omega_s$ , concluding the first statement of point (i) in the lemma.

As  $a \rightarrow 0$ , it holds that  $y_{min} \rightarrow 0$  and

$$\lim_{a \rightarrow 0} G(y_{min}, a) = \lim_{a \rightarrow 0} [2y_{min} - (y_{min} + b)] = -b < 0.$$

Also

$$\lim_{a \rightarrow \infty} G(y_{min}, a) = +\infty.$$

Therefore, together with

$$\frac{dG(y_{min}, a)}{da} = \frac{dy_{min}}{da} \frac{\partial G}{\partial y} + \frac{\partial G}{\partial a} > 0,$$

the above limits imply the existence of  $\bar{a}$ , such that, condition (E.11) holds if and only if  $a < \bar{a}$ . Given  $\omega_{u,0}$ , this result can be translated into  $\omega_s$  so that  $\tilde{\omega}_s = \frac{\omega_{u,0}}{\bar{a}}$ . Noting that  $a < \bar{a} \Leftrightarrow \omega_s > \tilde{\omega}_s$ , we conclude the second half of point (i) in the lemma.

Suppose that condition (E.11) holds, meaning that  $y_0$  and  $y_1$  exist. For  $j \in 0, 1$ ,  $\frac{\partial G}{\partial a} > 0$  implies that  $\frac{dy_0}{da} > 0$  and  $\frac{dy_1}{da} < 0$ . Also, the implicit function theorem leads to

$$\frac{dy_j}{da} = -\frac{1}{G^{(1)}(y_j, a)} \frac{\partial G(y_j, a)}{\partial a}.$$

By taking the total derivative of  $\theta_k$ ,

$$\begin{aligned}
\frac{d\theta_k}{da} &= \frac{1}{2} \frac{1}{\sqrt{y_j}} \frac{dy_j}{da} G^{(1)}(y_j, a) + \sqrt{y_j} \frac{\partial J^{(1)}(y_j, a)}{\partial a} \\
&= -\frac{1}{2} \frac{1}{\sqrt{y_j}} \left( 2y_j \frac{\partial J^{(1)}(y_j, a)}{\partial a} - \frac{\partial J(y_j, a)}{\partial a} \right) + \sqrt{y_j} \frac{\partial J^{(1)}(y_j, a)}{\partial a} \\
&= \frac{1}{2} \frac{1}{\sqrt{y_j}} \frac{\partial J(y_j, a)}{\partial a} \\
&< 0,
\end{aligned} \tag{E.14}$$

where the last inequality follows from (E.13). Therefore, we conclude that the boundary slopes,  $\theta_H$  and  $\theta_L$  are both increasing in  $\omega_s$ .

Consider the magnitude of the reactions. By plugging (E.13) into (E.14),

$$\left. \frac{d\theta_k}{da} \right| \propto \frac{\sqrt{y_j}}{(y_j + a)^2} \left( 1 + \frac{a}{y_j + b} \right) \left( 1 + \frac{1}{y_j} \frac{a^2}{a + b} \right).$$

It is straightforward that the last terms are all decreasing in  $y_j$ . Since we compare  $y_0 < y_1$  with  $a$  being fixed, we obtain  $\frac{d\theta_H}{d\omega_s} > \frac{d\theta_L}{d\omega_s}$ , thereby concluding the proof.

*Proof of Statements (iii) and (iv).* Note that  $\lim_{\omega_s \rightarrow \tilde{\omega}_s} y_j = y_{min}$ , and  $J(y_{min}) = K_{min}\beta_{min} = 2\beta_{min}J^{(1)}(y_{min})$  holds. Therefore, at this limit,

$$\theta'_H = \frac{J(\beta_{min}^2)}{2\beta_{min}} = \beta_{min}J^{(1)}(\beta_{min}^2) = \frac{K_{min}}{2} = \theta_k,$$

where the last equality holds because of (E.5). This concludes the proof of statement (iii).

Finally, consider the limit at  $\omega_s \rightarrow 0$ , which implies  $a \rightarrow \infty$  (given  $\omega_{u,0}$ ). Since  $y_{min} \searrow a$ , we can denote it as  $y_{min} = a + \epsilon$ , where  $\epsilon > 0$  is a small positive and  $\lim_{\omega_s \rightarrow \tilde{\omega}_s} \epsilon = 0$ . With this notation, function  $J$  is rewritten at the limit as

$$\lim_{\omega_s \rightarrow \tilde{\omega}_s} J(y_{min}, a) = \lim_{a \rightarrow \infty, \epsilon \rightarrow 0} \frac{\epsilon(2a + b + \epsilon)}{(2a + \epsilon)(a + b + \epsilon)} = 0.$$

Since  $\theta'_H = \frac{J(y_{min})}{2\sqrt{y_{min}}}$ , we conclude that  $\lim_{\omega_s \rightarrow \tilde{\omega}_s} \theta'_H = 0$ .

As for statement (iv), case 1 occurs if, and only if, the curve  $\theta = \frac{\varphi}{2} \sqrt{\frac{\omega_{u,1}}{\omega_s}}$  exceeds  $\theta_H$  at  $\omega_s = \tilde{\omega}_s$ . The above analyses have already established that  $\theta_H = \frac{K^{(1)}(y_{min})}{2}$ , where  $y_{min}$  is implicitly characterized by parameters  $a$  and  $b$ , we can define the threshold  $\varphi^*$  as

$$\varphi^* \equiv K^{(1)}(y_{min}) \sqrt{\frac{\omega_{u,0}}{\bar{a}\omega_{u,1}}}. \quad (\text{E.15})$$

□

Note that Proposition 3.1 follows from points (i) and (ii) of Lemma E.3, whereas Proposition ?? is shown by putting monotonically decreasing  $\theta$  in equation (??) together with points (i) and (ii).

## F Proof of Lemma ?? and Proposition ??

The optimal trading strategy satisfies the following condition.

$$\therefore (1 - 2\lambda_0\beta_0) + \frac{\psi}{2\lambda_1} \xi^2(\gamma^2\beta_0 + \gamma(1 - \gamma)) = 0. \quad (\text{F.1})$$

The marginal benefit of increasing  $s^*$  is given by

$$\mathbb{E}_\beta[v(\omega_s, \beta_0)] = \left[ (1 - \lambda_0\beta_0) + \frac{\psi}{4\lambda_1} \xi^2(\gamma^2\beta_0 + 2\gamma(1 - \gamma)) \right] \beta_0 + \frac{\psi}{4\lambda_1} \xi^2(1 - \gamma)^2 + (1 - \psi) \frac{1}{2} \sqrt{\frac{\omega_{u,1}}{\omega_s}}. \quad (\text{F.2})$$

Then, it holds that

$$\begin{aligned} \frac{\partial \mathbb{E}_\beta[v(\omega_s, \beta_0)]}{\partial \beta_0} &= \left[ (1 - 2\lambda_0\beta_0) + \frac{\psi}{2\lambda_1} \xi^2(\gamma^2\beta_0 + \gamma(1 - \gamma)) \right] \\ &+ \left[ -\beta_0 \frac{\partial \lambda_0}{\partial \beta_0} + \frac{\psi}{4\lambda_1} \frac{\partial}{\partial \beta_0} [\xi^2(\gamma^2\beta_0 + 2\gamma(1 - \gamma))] \right] \beta_0 + \frac{\psi}{4\lambda_1} \frac{\partial}{\partial \beta_0} (\xi^2(1 - \gamma)^2). \end{aligned}$$

Note that the first term is zero due to the envelope theorem using (F.1). By using the notations introduced in equations (E.1), the remaining terms are

summarized as follows.

$$-\beta_0 \frac{\partial}{\partial \beta_0} \frac{\beta_0}{\beta_0^2 + a} + \frac{\psi}{4\lambda_1} \left[ \beta_0^2 \frac{\partial}{\partial \beta_0} \left( \frac{\beta_0}{a + b + \beta_0^2} \right)^2 + 2\beta_0 \frac{\partial}{\partial \beta} \left( \frac{1}{a + b + \beta_0^2} \right)^2 b\beta_0 + \frac{\partial}{\partial \beta} \left( \frac{b}{a + b + \beta_0^2} \right)^2 \right]. \quad (\text{F.3})$$

From the proof in Appendix E, it holds that  $y > a$  in the equilibrium. Therefore, the first term is negative. As for the second term, it holds that

$$\begin{aligned} \text{Second terms of (F.3)} &= 2\beta_0^3 \left( \frac{a + b - \beta_0^2}{(a + b + \beta_0^2)^3} \right) + 2 \frac{a + b - 3\beta_0^2}{(a + b + \beta_0^2)^3} b\beta_0 - 4b^2 \frac{\beta_0}{(a + b + \beta_0^2)^3} \\ &\sim -y^2 + y(a + b - 3b) + b(a - b). \end{aligned}$$

Since  $y > a$ , the quadratic equation above is negative for all  $y > a$  if and only if it is negative at  $y = a$ . This is true because

$$-a^2 + a(a - 2b) + b(a - b) = -ab - b^2 < 0.$$

Therefore, the marginal benefit in (F.2) is decreasing in  $\beta_0$ , concluding the proof of Lemma ?? . Proposition ?? directly follows from Lemma ?? and equation (??).